# GENERAL SCIENCE GRADE 10: PLANT GROWTH AND DEVELOPMENT

*CBE Phenomenon-Driven Lesson Sequence — Sub-Strand 1.6 (8 Lessons)*

| SUB-STRAND OVERVIEW | |
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| **Grade Level** | 10 |
| **Subject** | General Science |
| **Strand** | Strand 1.0: Life Science |
| **Sub-Strand** | Sub-Strand 1.6: Plant Growth and Development |
| **Total Duration** | 8 lessons × 40 minutes = 320 minutes total |
| **Sub-Strand Content** | – Concept of growth and development in plants<br>– Seed dormancy<br>– Conditions necessary for germination<br>– Types of germination (epigeal and hypogeal)<br>– Primary and secondary growth in plants<br>– Role of growth hormones in plants<br>– Factors that contribute to growth and development in plants (water, temperature, light, nutrients) |
| **Learning Outcomes** | By the end of the sub-strand, the learner should be able to:<br>a) explain the meaning of growth and development in plants,<br>b) describe the causes of seed dormancy in plants,<br>c) investigate the conditions necessary for germination,<br>d) differentiate between epigeal and hypogeal germination,<br>e) distinguish between primary and secondary growth in plants,<br>f) describe factors that contribute to growth and development in plants,<br>g) describe the role of growth hormones in plants,<br>h) appreciate the concept of growth and development in plants. |
| **Core Competencies** | – Communication and Collaboration: as the learner shares information gathered on differences between plant growth and development, discussing findings from germination experiments with peers in plenary.<br>– Self-efficacy: as the learner works on practical activities to investigate epigeal and hypogeal germination using locally available seeds such as maize (hypogeal) and beans (epigeal).<br>– Learning to Learn: as the learner reflects on the experiments investigating conditions necessary for germination and adjusts conclusions based on evidence gathered. |
| **Core Values** | – Responsibility: Learner engages in assigned roles while working with others in the process of learning such as investigating conditions necessary for germination.<br>– Integrity: as the learner records and reports experimental observations honestly, without altering data from germination and growth investigations. |

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| | – Patriotism: as the learner connects knowledge of plant growth and development to improving crop yields and food security in Kenya (e.g., maize and bean farming in the Rift Valley). |
| **Science & Engineering Practices** | – Planning and Carrying Out Investigations: as the learner designs and conducts experiments to investigate conditions necessary for germination (water, air, warmth) and compares epigeal vs. hypogeal germination in maize and bean seeds.<br>– Analyzing and Interpreting Data: as the learner measures and records seedling growth over time, interprets results to distinguish primary from secondary growth, and draws evidence-based conclusions.<br>– Constructing Explanations: as the learner uses gathered evidence and knowledge of growth hormones (auxins, gibberellins) to explain observed plant growth patterns.<br>– Obtaining, Evaluating, and Communicating Information: as the learner uses digital and print media to research growth hormones and seed dormancy, then shares findings with peers using presentations. |
| **Pertinent & Contemporary Issues (PCIs)** | – Cyber Security: as the learner observes online safety when searching for information on growth and development of plants using digital devices.<br>– Environmental Issues: as the learner connects understanding of plant growth factors (water, nutrients, light) to sustainable farming practices and food security in Kenya.<br>– Financial Literacy: as the learner explores how knowledge of seed dormancy and germination conditions can improve subsistence and commercial crop production (e.g., maize, wheat, and tea farming in Kericho and the Rift Valley). |
| **Career Connections** | – Agronomist / Crop Scientist: understanding plant growth, development, and hormones underpins careers in crop improvement and large-scale farming across Kenya's agricultural regions.<br>– Horticulturalist: knowledge of germination, primary and secondary growth, and growth hormones is essential for flower and vegetable farming, including Kenya's export horticulture sector in Naivasha and Thika.<br>– Botanist / Plant Biologist: investigating plant growth and development forms the foundation of botanical research and biodiversity conservation in national parks and forests.<br>– Seed Technologist: understanding seed dormancy and the conditions necessary for germination is directly applied in seed processing, storage, and quality testing industries.<br>– Agricultural Extension Officer: communicates knowledge of plant growth factors and germination to smallholder farmers to improve food crop yields across rural Kenya. |
| **Focus for Lessons** | This unit explores how plants grow and develop from seed germination through to secondary growth, anchored in the puzzling observation that identical maize and bean seeds planted at the same time and place can emerge, grow, and develop in strikingly different ways. Learners investigate seed dormancy, germination conditions, types of germination, growth hormones, and primary versus secondary growth through hands-on experiments using locally available seeds and plants. The unit emphasises evidence-based reasoning, collaborative inquiry, and connections to Kenyan agriculture and food security. |
| **Driving Question / Key Inquiry Questions** | DRIVING QUESTION: Why do seeds from the same garden grow and develop so differently, and what controls how a plant grows from a tiny seed into a tall tree or a flowering herb?<br><br>KICD KEY INQUIRY QUESTION:<br>1. Why is the concept of growth in plants important? |
| **Anchoring Phenomenon** | A farmer in Kisumu plants two handfuls of seeds — one handful of maize and one handful of beans — in the same garden bed on the same day, using the same soil, and watering them equally. Within a week, striking differences are visible: the bean seedlings push their seed leaves (cotyledons) above the soil surface and unfurl into a distinct arch shape, while the maize seedlings shoot a single green spike straight upward with the seed remaining underground. As weeks pass, the bean plants grow taller and branch outward, while nearby mango and avocado trees — planted years earlier — show thickening, woody trunks and spreading |

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| | canopies. Some seeds from a packet stored for two seasons fail to germinate at all, even when given water and warmth. Students observe these differences and are puzzled: Why do maize and bean seeds emerge so differently? Why do some seeds refuse to germinate even when given the right conditions? And why do some plants remain thin and herbaceous while others become thick and woody over time? |
| **Storyline Thread** | Lesson 1 — Noticing and Wondering (Anchoring Phenomenon + Predict Phase): Students observe the anchoring phenomenon — side-by-side germination of maize and bean seeds in transparent containers, plus photographs of a mango tree trunk cross-section and a wilted, dormant seed packet. Students record predictions about why the seeds look and behave differently, and co-construct an initial Driving Question Board (DQB) with "What we think we know" and "What we wonder" columns. Key vocabulary introduced: growth vs. development.<br><br>Lesson 2 — What Is Dormancy? (Seed Dormancy + Observe Phase): Students use print and digital media to investigate the causes of seed dormancy (hard seed coat, immature embryo, chemical inhibitors, unfavourable environmental conditions). They examine dormant and non-dormant seeds and discuss methods of breaking dormancy (scarification, stratification, soaking). Kenyan context: sorghum and bean seed storage by farmers in Kisumu and Turkana. Students update the DQB with new questions about why some seeds refuse to germinate.<br><br>Lesson 3 — What Does a Seed Need? (Conditions for Germination — Experiment): Students plan and carry out a structured experiment to investigate the conditions necessary for germination (water, air/oxygen, optimum temperature) using maize or bean seeds, setting up control and experimental groups. They record observations daily and begin an evidence table. NGSS SEP: Planning and Carrying Out Investigations. Students revisit predictions from Lesson 1 and note surprises.<br><br>Lesson 4 — Two Ways to Emerge (Types of Germination — Observe + Explain Phase): Students observe and compare epigeal germination (beans — cotyledons emerge above soil) and hypogeal germination (maize — cotyledons remain underground) using germinating seeds in transparent containers. They draw and label diagrams of each type, discuss differences in cotyledon position and function, and explain why the differences exist. Connection to the anchoring phenomenon: the Kisumu farmer's puzzling observation is now explained. DQB updated.<br><br>Lesson 5 — Getting Taller vs. Getting Thicker (Primary and Secondary Growth — Explain Phase): Students distinguish between primary growth (apical meristems — increase in length) and secondary growth (lateral meristems / cambium — increase in girth/width). They examine cross-sections of a herbaceous stem (sukuma wiki) vs. a woody stem (mango or avocado twig) and identify vascular cambium. Supporting phenomenon: the mango tree trunk that widens every year. Students build an initial explanatory model of how a tree trunk grows thicker.<br><br>Lesson 6 — Chemical Messengers in Plants (Growth Hormones — Observe + Explain Phase): Students use print and digital media to research the role of growth hormones — auxins (phototropism, apical dominance), gibberellins (stem elongation, seed germination), cytokinins (cell division), abscisic acid (dormancy), and ethylene (fruit ripening). They connect the spindly sukuma wiki in the dark (supporting phenomenon) to auxin redistribution. Students design a simple demonstration of phototropism using a potted seedling and a light source (if available). DQB updated with hormone connections.<br><br>Lesson 7 — Factors Favouring Growth (Environmental Factors + Model Building): Students investigate and discuss how water, temperature, light, and nutrients influence plant growth and development through a structured group activity. Each group analyses a case study of crop growth in different Kenyan regions (tea in Kericho — cool and wet; maize in the Rift Valley — warm with seasonal rain; mangroves in Mombasa — saline and humid). Groups present findings and the class builds a cumulative explanatory model connecting germination conditions, growth types, hormones, and environmental factors to the anchoring |

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| | phenomenon.<br><br>Lesson 8 — Putting It All Together (Synthesis + Cumulative Model Revision + Assessment): Students revise their individual and class models to produce a complete, labelled explanatory diagram that traces a seed's journey from dormancy through germination, primary growth, secondary growth, and the influence of hormones and environmental factors. They revisit the DQB and mark all answered questions. A structured performance task asks students to explain in writing or orally: "A farmer in Nakuru notices her bean seedlings are growing tall and pale instead of short and bushy. Using what you know about plant growth and development, explain what might be happening and what she could do." Class closes by connecting learning to careers in agronomy, horticulture, and seed technology in Kenya. |
| **Supporting Phenomena** | – A mango tree in a Mombasa homestead has a thick, woody trunk that widens each year, while a nearby tomato plant remains soft-stemmed and dies after one season — what causes this difference in growth pattern?<br>– A farmer in the Rift Valley stores sorghum seeds in a granary for six months; when planted, some germinate readily while others remain dormant for weeks — what determines whether a seed germinates or stays dormant?<br>– A young eucalyptus tree in Nairobi is cut near the base, yet new shoots sprout from the stump within weeks, growing rapidly upward — what internal signals drive this regrowth?<br>– Sukuma wiki (kale) seedlings grown in a dark room become tall, pale, and spindly, while those grown in sunlight are short, dark green, and sturdy — how does light influence the direction and quality of plant growth? |

| LESSON 1 (40 minutes): Noticing and Wondering: Why Do Seeds Behave So Differently? | |
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| **A. SPECIFIC LEARNING OUTCOMES** | |
| **Category** | **Specific Learning Outcomes** |
| **Purpose** | Students will anchor their curiosity in a real Kenyan farming observation and begin building questions that will drive the entire sub-strand investigation. |
| **Knowledge** | Students will distinguish between the terms growth (an increase in size or mass) and development (a series of organised changes in structure and function) as applied to plants. |
| **Skills** | Students will make and record initial predictions about seed behaviour, observe physical differences between maize and bean germination setups, and co-construct a Driving Question Board. |
| **Attitudes** | Students will demonstrate curiosity and openness by sharing initial ideas without fear of being wrong, valuing all contributions to the class question board. |
| **Key Inquiry Question** | Why is the concept of growth in plants important? |
| **Purpose in Storyline** | This is the anchoring lesson that launches the phenomenon. Students observe striking differences in maize and bean germination side by side, encounter the puzzle of non-germinating seeds, and generate the questions that the next seven lessons will answer. The Driving Question Board created here will be revisited and updated in every subsequent lesson. |
| **Safety Notes** | Ensure transparent containers used for germination setups are stable and will not tip. If seeds have been pre-soaked, remind students to wash hands after handling them. No chemical hazards in this lesson. |

## B. LESSON OVERVIEW

The teacher introduces the anchoring phenomenon through a narrated scenario about a Kisumu farmer who plants maize and beans side by side and notices puzzling differences. Students observe live or prepared transparent germination containers showing maize and bean seeds at day 3 to 5 of germination, plus printed photographs of a mango tree trunk cross-section and a wilted seed packet labelled as two seasons old. Students record individual predictions, share with partners, then contribute questions and initial ideas to a class Driving Question Board divided into two columns: What we think we know and What we wonder. The lesson closes by formally introducing the vocabulary distinction between growth and development.

## C. LESSON IMPLEMENTATION FRAMEWORK

| Phase | Learner Experience | Teacher Moves | Sensemaking Strategy | Formative Assessment Strategy |
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| **Predict Phase**<br>📹 **VIDEO:** [Organism growth and the environment]<br>Source: Khan Academy (English | Students listen to the teacher narrate the Kisumu farmer scenario without any explanation given. They are then shown the two transparent germination containers and the photographs. Each student silently writes in their | Set the scene slowly and with detail: Imagine you are Akinyi, a smallholder farmer near Kisumu. One Monday morning she takes two handfuls of seeds from her store — one handful of maize, one of beans — and plants them in the | Think-Write-Pair-Share anchored to observable evidence in the containers and photographs. Students connect their everyday knowledge of seeds and farming to a novel observation, activating prior knowledge about germination | Teacher circulates during silent writing and notes two or three student ideas to highlight — particularly any student who mentions soil, water, the type of seed, or the position of the seed leaf. These ideas will be used to |

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| - US curriculum)<br>🔍 Search ARES for similar videos<br><br>📖 **HTML:**<br>Exponential Growth and Decay (Flexbook)<br>Source: CK-12<br>🔍 Search ARES for similar readings | exercise book answers to three prompts: (1) What do you notice that is different between the maize and the bean setups? (2) Why do you think the seeds are behaving differently? (3) What do you think will happen to each plant in the next few weeks? Students have 4 minutes of individual thinking time before any sharing begins. | same bed, covers them with the same soil, and waters them the same amount. By Saturday she walks out and sees something that confuses her. Hold up or point to the containers and photographs. Then say: Before we talk to anyone, I want each of you to write what YOU notice and what YOU think is happening. Do not worry about being right. We are scientists starting an investigation. After issuing the three written prompts, enforce 4 minutes of strict silent individual writing. Use a visible timer. Walk around and read over shoulders without commenting. After 4 minutes say: Now turn to your partner and share your three answers. You each have 90 seconds. I will listen. WAIT TIME NOTE: After the pair-share, pause 10 seconds before calling on any student to allow all pairs to finish a thought. | without the teacher providing any explanatory content yet. | seed the DQB. Teacher checks that all students have written at least one observation and one prediction before pair-share begins. |
| **Observe Phase**<br>📹 **VIDEO:**<br>Organism growth and the environment<br>Source: Khan Academy (English - US curriculum)<br>🔍 Search ARES for similar videos<br><br>📖 **HTML:**<br>Exponential Growth and Decay (Flexbook)<br>Source: CK-12<br>🔍 Search ARES for similar readings | Pairs are invited to come forward in rotation (or containers are passed around if space is limited) to observe the germination containers up close for 2 minutes per pair or group. Students are given a simple observation guide with four boxes: (1) Sketch what you see in the maize container, (2) Sketch what you see in the bean container, (3) Circle any differences you see, (4) Write one question the observation makes you want to ask. Students also examine the two photographs — the mango trunk cross-section showing rings and the wilted seed packet — and add observations about those to their guides. | As students observe, circulate and use probing questions without giving answers: Where is the seed in each container? Where are the leaves coming from in the bean setup compared to the maize setup? What do you think is inside the mango trunk that makes those rings? What might have happened to the seeds in that old packet? Do not confirm or deny any student idea. When all groups have observed, bring the class back together and say: I heard some really interesting noticings. Let us share a few — but remember, we are not explaining yet. We are only describing what we observe. Call on 4 to 5 students. After each student shares, ask the class: Does anyone else notice something different or something | Structured observation guide forces students to separate observation from explanation — a key scientific practice. Sketching builds visual literacy and gives students a record to compare against later lessons when they learn the formal terms epigeal and hypogeal germination. | Teacher scans completed observation guides for evidence that students can distinguish between the two germination patterns. Look specifically for whether students note cotyledon position above vs. below soil surface even if they do not yet know the vocabulary. Students who only note colour or size differences will be prompted with: Look carefully at where the seed itself is in each container. |

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| | | to add? WAIT TIME NOTE: After asking Does anyone else notice something different, wait 8 seconds in silence before accepting a response. This silence communicates that observation takes time. | | |
| **Explain Phase**<br>📹 **VIDEO:** Organism growth and the environment<br>Source: Khan Academy (English - US curriculum)<br>🔍 Search ARES for similar videos<br><br>📖 **HTML:** Exponential Growth and Decay (Flexbook)<br>Source: CK-12<br>🔍 Search ARES for similar readings | Students attempt a first explanation in writing. The prompt displayed on the board reads: Using only what you have observed so far — not what you have read or been taught — write two or three sentences explaining why you think maize and bean seeds emerge so differently. Students write individually for 3 minutes. A volunteer shares their explanation aloud. The teacher then introduces exactly two vocabulary terms — growth and development — using a simple anchor definition chart written on the board: Growth means an increase in size or mass. Development means a series of organised changes in structure and function, like a seed becoming a seedling and then a tree. The teacher gives a Kenyan example: When a mango seedling in a nursery in Thika gets taller, that is growth. When the same mango seedling grows its first true leaves and then later forms flowers, that is development. Students copy the definitions and add one example of each from their own observation. | After students write their first explanations say: These are your starting ideas. Scientists call these initial models. They are meant to be changed as we learn more. We will come back to what you wrote today in Lesson 8 and you will be amazed at how much your thinking has grown. Introduce the two vocabulary terms without lecturing. Write them large on the board. Say: Growth and development are two words people often use as if they mean the same thing. In plant science, they mean different things. Let me give you a way to remember. Growth is your mango tree getting taller — you can measure it with a ruler. Development is that same mango tree producing a flower for the first time — something changed in what it can DO. Ask: Can you now give me an example of growth from our containers? And an example of development? WAIT TIME NOTE: After each question, wait 6 seconds before accepting answers. If no response after 6 seconds, say: Talk to your partner for 30 seconds, then I will ask again. | Initial explanation writing followed by formal vocabulary introduction ensures that new terms are attached to observations students have already made, not introduced in the abstract. The Thika mango nursery example connects the distinction to a familiar Kenyan agricultural context. | Teacher reads two or three student initial explanations aloud (with permission or anonymously) and asks the class: Does this explanation fit what we observed? What would we need to know to decide if it is correct? This surfaces gaps in reasoning without correcting students, building a culture of evidence-based thinking. |
| **Driving Question Board (DQB) Creation**<br>📹 **VIDEO:** Organism growth and the | Each student is given two sticky notes or small paper slips of different colours — one for What we think we know and one for What we wonder. Students write one idea or fact they believe is true about plant growth on the first slip, | Say: We are going to build something that will live on our classroom wall for the next eight lessons. It is called a Driving Question Board. Every time we learn something new, we will come back and update it. Right now, I | The DQB externalises student thinking and makes it a community resource. Colour-coded slips (claims vs. questions) train students to distinguish between what they know and what they are uncertain about — a foundational | Teacher photographs the completed DQB. Review of the What we think we know slips reveals prior conceptions — including possible misconceptions such as plants grow because of soil or bigger seeds grow into |

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| environment<br>Source: Khan Academy (English - US curriculum)<br>🔍 Search ARES for similar videos<br><br>📖 **HTML:** Exponential Growth and Decay (Flexbook)<br>Source: CK-12<br>🔍 Search ARES for similar readings | and one question they genuinely want answered on the second slip. Students post their slips on a class chart divided into two columns. The teacher reads several aloud and groups similar questions together on the board. The class agrees on a shared wording for the top three to five most important questions. The Driving Question is written at the top of the board: Why do seeds from the same garden grow and develop so differently, and what controls how a plant grows from a tiny seed into a tall tree or a flowering herb? | want to know what YOU think and what YOU wonder — not what your textbook says. After students post their slips, read several aloud with energy and genuine curiosity. Say things like: Oh, this one is fascinating — someone wants to know why old seeds stop working. We do not know the answer yet, but we will. After grouping questions, say: Look at this board. All of these questions came from YOUR observations today of a simple farming scenario. This is exactly how real scientists in Kenya — like those at KALRO, the Kenya Agricultural and Livestock Research Organisation — start their research. They notice something in the field and ask why. WAIT TIME NOTE: When asking students to share which question they most want answered, pause 10 seconds after the question is posed to allow genuine reflection before taking hands. | epistemic practice. Connecting the practice to KALRO researchers grounds scientific inquiry in a Kenyan institutional context. | bigger plants — that will need to be addressed across the sub-strand. Review of the What we wonder slips informs lesson pacing and emphasis for Lessons 2 through 8. |
| **Model Building Phase**<br>📹 **VIDEO:** Organism growth and the environment<br>Source: Khan Academy (English - US curriculum)<br>🔍 Search ARES for similar videos<br><br>📖 **HTML:** Exponential Growth and Decay (Flexbook)<br>Source: CK-12<br>🔍 Search ARES for similar readings | Students draw a simple initial model in their exercise books — a blank outline of a seed and a question mark above it. Inside the seed they write what they think is inside a seed that determines how it grows. Around the outside they draw arrows and label the conditions they think a seed needs. This is explicitly framed as a first draft that will be wrong in places and will be revised many times. Students write their name and the date on the model and label it Model Version 1. The teacher collects or photographs these models for comparison in Lesson 8. | Say: I want you to draw your thinking, not a perfect diagram. You do not need labels you cannot spell. You do not need to know the right answers. I want to see what is inside your head right now about how a seed becomes a plant. Give 4 minutes for drawing. Then say: Look at your model. Put a star next to any part you are confident about. Put a question mark next to any part you are unsure of. In Lesson 8, we will look at this same model and you will revise it using everything we have discovered together. That revision is part of your assessment. After students mark their models say: This is how science works. No scientist starts with a perfect answer. They start | Initial model drawing serves as both a formative assessment tool and a motivational anchor — students are invested in their own model and will want to see how it changes. The star and question mark annotation builds metacognitive awareness about certainty vs. uncertainty, a key aspect of scientific epistemology. | Teacher collects or photographs Model Version 1 from every student. Before the next lesson, review models to identify (1) students who have no idea what is inside a seed — these students may need additional scaffolding in Lesson 2 around seed structure, and (2) students who already know terms like embryo or endosperm — these students can be given extension roles such as peer explainers or group leads in Lesson 3. |

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| | | with the best idea they have, test it, and improve it. Your model today is the start of your scientific thinking about plant growth and development. WAIT TIME NOTE: After issuing the model drawing task, do not speak for the full 4 minutes. Silence during model building signals that this is genuine thinking time, not a quick task. | | |

| D. TEACHER REFLECTION |
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| After the lesson, ask yourself: Did every student contribute at least one sticky note to the DQB? If not, who was silent and why? Were the germination containers set up early enough to show visible differences — ideally day 3 to 5? If the bean cotyledons are not yet above soil, the visual contrast is lost. Did students distinguish between observing and explaining, or did they jump to conclusions? Were there any student questions on the DQB that I had not anticipated and that I need to research before Lesson 2? Did the Kisumu farmer scenario feel real and relatable to students, or did it feel distant? If students seemed disengaged with the narrative, consider asking if any student has a family member who farms and inviting them to briefly describe what they have seen. |

| E. SUMMARY TABLE PROMPT (pre-filled example for this lesson) | |
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| **What did I observe?** | Students observed two transparent germination containers showing maize and bean seeds at day 3 to 5 of growth, noting that bean cotyledons arch above the soil while maize sends a single spike upward with the seed remaining underground. They also examined a photograph of a mango trunk cross-section showing growth rings and a photograph of a wilted two-season-old seed packet. |
| **What did I learn?** | Students learned the distinction between growth — an increase in size or mass — and development — a series of organised changes in structure and function. They learned that scientists begin investigations by making careful observations and recording honest questions, as researchers at institutions like KALRO in Kenya do. |
| **How does this explain the phenomenon?** | Students produced initial explanations for why maize and bean seeds emerge differently, drawing on their own prior knowledge and observations. These explanations are first drafts and are expected to be incomplete. Students also built Model Version 1 showing their current understanding of what is inside a seed and what it needs to grow, which will be revised across the sub-strand. |

| LESSON 2 (40 minutes): What Is Dormancy? Why Do Some Seeds Refuse to Wake Up? | |
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| **A. SPECIFIC LEARNING OUTCOMES** | |
| **Category** | **Specific Learning Outcomes** |
| **Purpose** | Students will investigate the causes of seed dormancy and the methods used to break it, connecting their findings to the puzzling observation from Lesson 1 about seeds that fail to germinate even when given water and warmth. |
| **Knowledge** | Students will be able to: (1) define seed dormancy as a state in which a viable seed does not germinate even under conditions that would normally support growth; (2) identify four causes of dormancy: hard or impermeable seed coat, immature or undeveloped embryo, presence of chemical inhibitors such as abscisic acid, and unfavourable environmental conditions such as lack of sufficient light or temperature; (3) describe three methods of breaking dormancy: scarification (physically or chemically abrading the seed coat), stratification (exposing seeds to cold and moist conditions), and soaking in water. |
| **Skills** | Students will practise: (1) close observation and comparison of dormant versus non-dormant seeds using physical examination; (2) extracting and synthesising information from print and digital media sources; (3) scientific reasoning by linking evidence about seed structure to explanations of dormancy. |
| **Attitudes** | Students will appreciate that dormancy is an adaptive advantage that has allowed farmers across Kenya to store seeds across seasons, and they will develop curiosity about how traditional seed-keeping practices in Kisumu and Turkana connect to the science of dormancy. |
| **Key Inquiry Question** | Why is the concept of growth in plants important? This lesson advances the inquiry by showing that growth cannot begin until dormancy is broken, making dormancy a critical gatekeeper of the entire growth and development process. |
| **Purpose in Storyline** | This is Lesson 2 of 8. In Lesson 1 students anchored their learning in the Kisumu farmer phenomenon and noticed that some seeds from a stored packet failed to germinate at all. This lesson zooms in on that specific puzzle — the dormant seeds — and builds the first layer of scientific explanation. By understanding dormancy now, students will have a foundation for understanding what conditions seeds actually need to germinate in Lesson 3, and why maize and bean emerge differently in Lesson 4. |
| **Safety Notes** | No significant hazards. If razor blades or craft knives are used to cut open seeds for observation, teacher should supervise closely and remind students to always cut away from fingers. Soaked seeds left in warm conditions for more than 24 hours may develop mould — students should wash hands after handling any seeds that show mould growth. |

## B. LESSON OVERVIEW

Students begin by revisiting their Lesson 1 predictions about why the stored seeds in the Kisumu farmer phenomenon failed to germinate. They then examine pairs of dormant and non-dormant seeds — using sorghum, bean, or maize from locally sourced packets — observing differences in seed coat texture, colour, and firmness. Using print resources, a short video clip (where internet is available), or teacher-guided notes, students investigate the four main causes of dormancy and three methods of breaking it. The lesson draws explicitly on Kenyan farmer practices: how sorghum growers in Turkana county dry and store seeds to maintain dormancy across dry seasons, and how bean farmers around Kisumu sometimes nick or soak seeds before planting. The class closes by adding new, more precise questions to the Driving Question Board, replacing vague wonders with evidence-informed questions.

## C. LESSON IMPLEMENTATION FRAMEWORK

| Phase | Learner Experience | Teacher Moves | Sensemaking Strategy | Formative Assessment Strategy |
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| **Predict Phase**<br>📹 **VIDEO:**<br>Organism growth and the environment<br>Source: Khan Academy (English - US curriculum)<br>🔍 Search ARES for similar videos<br><br>📖 **HTML:**<br>Seeds and Seed Dispersal (Flexbook)<br>Source: CK-12<br>🔍 Search ARES for similar readings | Students open their science notebooks to their Lesson 1 DQB entries. The teacher displays or reads aloud one specific student question recorded in the previous lesson — chosen to be a question about why the stored seeds would not germinate. Students spend 3 minutes writing a personal prediction in response to the prompt: 'A farmer in Kisumu stored bean seeds in a dry gourd for two seasons. She plants them with water and warmth but they do not sprout. Write two possible reasons why.' Students write independently without consulting neighbours, committing their current thinking to paper. After writing, two or three volunteers share their predictions aloud. The teacher records these on the board in a column labelled OUR PREDICTIONS without correcting or affirming — all ideas are treated as hypotheses to be tested against evidence. | Begin with: 'Last lesson we noticed something strange in our farmer story — she gave the seeds water and warmth but some still did not grow. You wrote questions about this on our DQB. Today we are going to investigate that specific mystery.' Display the prompt on the board. After students write, use cold-calling with a warm tone: 'Akinyi, what did you predict? Let us hear your thinking.' After each response, say: 'Interesting. Does anyone have a different idea?' WAIT 8 SECONDS before moving on. Do not say correct or incorrect. Write all predictions on the board. Close the predict phase by saying: 'We now have competing ideas. Scientists call this having multiple hypotheses. Our job today is to find evidence that either supports or challenges what we have written here.' | Activating prior knowledge — students surface existing mental models about seeds and germination. The teacher deliberately keeps all predictions visible throughout the lesson so students can return to them later and self-correct or confirm. | Teacher reads students written predictions as they circulate during the 3-minute writing time. Look for: (1) students who confuse a dead seed with a dormant seed — a key misconception to address; (2) students who already mention the seed coat or storage conditions, showing prior knowledge that can anchor peer discussion. Note names for targeted questioning during the explain phase. |
| **Observe Phase**<br>📹 **VIDEO:**<br>Organism growth and the environment<br>Source: Khan Academy (English - US curriculum)<br>🔍 Search ARES for similar videos<br><br>📖 **HTML:**<br>Seeds and Seed Dispersal (Flexbook)<br>Source: CK-12<br>🔍 Search ARES for similar readings | Students work in pairs. Each pair receives a small tray with four seeds: (1) a freshly harvested or recently purchased bean seed with intact testa, (2) a bean seed that has been soaked overnight in water, (3) a sorghum seed from a commercial packet, and (4) a sorghum seed from a packet labelled or simulated as two seasons old. If actual old seeds are unavailable, the teacher can simulate by preparing seeds that were dried in an oven at low heat for 30 minutes — this does not kill all seeds but reduces germination rate noticeably, replicating the phenomenon. Students use a simple observation guide with the | Distribute materials and say: 'You have 7 minutes. Do not crack or split the seeds yet — just observe the outside. Use your fingers as your first instrument. Ask yourself: does this seed feel like it is ready to do something, or does it feel locked up?' Circulate and ask probing questions: 'What does the texture tell you about what might be happening inside?' and 'If you were a tiny root trying to push out of this seed, what would you have to get through?' WAIT 5 SECONDS after each question before offering any guidance. If students finish early, ask: 'Can you gently try to bend the soaked seed versus the dry seed? What do you | Hands-on sensory observation builds concrete referents for abstract vocabulary. By feeling the difference between a hard dry seed coat and a softened soaked one, students develop embodied understanding of what a seed coat does and why it matters for dormancy. This also connects to the Kisumu farmer context — students can imagine a farmer feeling seeds to decide if they are ready to plant. | Circulate and check observation tables. Target students who are writing only superficial notes such as 'the seeds are brown.' Ask: 'Can you be more precise? How does the coat of seed one compare to seed two when you press gently?' This pushes toward comparative scientific language. Students who note that soaked seeds feel swollen or flexible have correctly observed a key indicator of dormancy breaking. |

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| | following columns: seed type, colour, texture of coat, flexibility when pressed gently, and any visible change. They spend 7 minutes examining the seeds with their fingers and, where available, a hand lens. Students record in their notebooks. After individual observation, pairs join to form groups of four and compare findings. Each group nominates a recorder who lists the differences they all agreed on. | notice?' After group comparison, ask one group to share: 'Group three, what was the most surprising difference you found?' WAIT 8 SECONDS. | | |
| **Explain Phase**<br>📹 **VIDEO:**<br>Organism growth and the environment<br>Source: Khan Academy (English - US curriculum)<br>🔍 Search ARES for similar videos<br><br>📖 **HTML:**<br>Seeds and Seed Dispersal (Flexbook)<br>Source: CK-12<br>🔍 Search ARES for similar readings | The teacher leads a 12-minute structured information-processing activity. Where internet is available, students watch a 3-minute curated video clip showing seed dormancy and germination in slow motion (KALRO educational content or similar Kenyan agricultural extension materials where accessible). Where internet is unavailable, the teacher provides a one-page illustrated reading with diagrams of the four causes of dormancy and three methods of breaking it. Students read independently for 4 minutes, then the teacher leads a class discussion using a graphic organiser drawn on the board — a two-column table: CAUSE OF DORMANCY on the left and WHAT IT MEANS FOR THE SEED on the right. Students contribute to filling in the table as a class. The four causes addressed are: (1) hard or impermeable seed coat — water and gases cannot enter; (2) immature embryo — the seed needs more time to develop fully before it can germinate; (3) chemical inhibitors such as abscisic acid — the seed produces substances that actively prevent germination until conditions | Before the reading or video, say: 'As you read or watch, I want you to hold one question in your mind: which of these causes best explains why the farmer in Kisumu stored seeds did not germinate? Underline or mark anything that helps you answer that question.' After the activity, say: 'Let us build our cause-and-effect table together. Who found something that explains what stops a seed from germinating?' Call on students who were quiet during the predict phase to give them a second entry point. When introducing the Kenyan context, say: 'A sorghum farmer in Turkana is already a seed scientist — she has figured out through practice what researchers at KALRO study in laboratories. What does that tell us about the relationship between traditional knowledge and science?' WAIT 10 SECONDS — this is a high-level question and deserves extended thinking time. Do not answer it yourself; instead, reflect back what students say: 'So you are saying farmers discovered this by observation over generations — that is exactly how science works too.' | The cause-and-effect graphic organiser helps students organise new vocabulary into a structure rather than a list. Connecting Turkana and Kisumu farming practices to the scientific concepts honours indigenous knowledge systems and helps students see science as relevant to their community. The teacher deliberately positions traditional farmer knowledge as scientifically valid, which supports positive scientific identity formation. | As students copy the table, circulate and check for the key conceptual distinction: dormancy is not the same as death. Ask any student whose table conflates the two: 'Is a dormant seed alive or dead? How do you know from what we just read?' Students should be able to say: a dormant seed is alive but its metabolism is slowed down and germination is blocked. This is a critical concept for Lesson 3. |

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| | change; (4) unfavourable environmental conditions — the seed is waiting for the right temperature, season, or light signal. The teacher then introduces three methods of breaking dormancy: scarification, stratification, and soaking. The Kenyan context is woven in explicitly: sorghum farmers in Turkana dry and store seeds across dry seasons to keep them dormant until the rains arrive, and bean farmers around Kisumu sometimes nick the testa with their thumbnail or soak seeds overnight — both are practical forms of scarification and soaking. Students copy the completed table into their notebooks. | | | |
| **Driving Question Board (DQB) Creation**<br>📹 **VIDEO:**<br>Organism growth and the environment<br>Source: Khan Academy (English - US curriculum)<br>🔍 Search ARES for similar videos<br><br>📖 **HTML:**<br>Seeds and Seed Dispersal (Flexbook)<br>Source: CK-12<br>🔍 Search ARES for similar readings | Students return to the class DQB from Lesson 1, which has two columns: WHAT WE THINK WE KNOW and WHAT WE WONDER. The teacher displays it and reads out the questions already posted. Students work individually for 2 minutes to write one new question that today's lesson has raised for them on a sticky note or small card — the question must be more precise than anything on the DQB already. Examples of strong new questions students might generate: 'If abscisic acid stops germination, what chemical breaks it down?', 'How does a seed know when the rainy season has started?', or 'Can all seeds be scarified or do some get damaged?' Students post their new questions on the DQB in the WHAT WE WONDER column. The teacher reads two or three aloud and says: 'These are the kinds of questions that plant scientists in Kenya ask. Some of these we will | Point to the original DQB and say: 'Before today, we wondered in a general way why some seeds do not germinate. Now we have some evidence. I want to see if your questions have become more precise — more like a scientist would ask them.' After students post new questions, select one vague original question from Lesson 1 and one precise new question from today. Hold them up side by side and ask: 'What is the difference between these two questions? Which one would be easier for a scientist to investigate, and why?' WAIT 8 SECONDS. Guide students to see that precise, testable questions are more useful for investigation. Say: 'As we learn more each lesson, your questions should keep getting sharper. That is a sign of real scientific thinking.' | The DQB update serves as a metacognitive tool — students can see their own thinking becoming more sophisticated. Comparing a vague Lesson 1 question to a precise Lesson 2 question makes the growth in scientific thinking visible and celebrates it, building confidence and intellectual identity. | Collect the sticky notes or read new questions as students post them. Look for: (1) questions that are testable and specific — these show deep engagement; (2) questions that reveal the misconception that dormancy equals death — use these to plan a brief correction at the start of Lesson 3; (3) students who cannot generate a new question — sit with them briefly and ask: 'What confused you today? Write that confusion as a question.' Confusion is valid scientific wondering. |

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| | answer in the next lessons. Some are questions that researchers are still investigating.' | | | |
| **Model Building Phase**<br>📹 **VIDEO:** [Organism growth and the environment]<br>Source: Khan Academy (English - US curriculum)<br>🔍 Search ARES for similar videos<br><br>📖 **HTML:** [Seeds and Seed Dispersal (Flexbook)]<br>Source: CK-12<br>🔍 Search ARES for similar readings | In the final 4 minutes, students make the first addition to their individual explanatory model — a simple annotated sketch begun in Lesson 1. Students draw a seed and label two things: (1) one cause of dormancy visible on or in the seed, and (2) one arrow showing what happens when that cause is removed. For example, a student might draw a seed with a thick testa labelled 'hard seed coat blocks water entry' and an arrow after scarification labelled 'water enters, embryo activates.' This is not a polished diagram — it is a working model. The teacher emphasises: 'Your model will change every lesson. Today you are adding just one idea to it. By Lesson 8 it will tell the whole story of a plant from dormant seed to tall tree.' | Say: 'Open your model from last lesson — the sketch you started. We are going to add one idea today. Just one. Do not try to draw everything we learned — pick the cause of dormancy that makes the most sense to you and show what happens when it is removed.' Circulate and ask: 'Why did you choose that cause to draw?' WAIT 5 SECONDS. Encourage students who are drawing only and not labelling: 'Your drawing is great — now tell the story with words too. Add a label that explains the arrow.' Close the lesson by saying: 'Next lesson we are going to ask: even when dormancy is broken, what exact conditions does a seed need to germinate? You are going to design an experiment to find out. Come ready to be scientists.' | The model-building phase makes student thinking visible in a form that accumulates across all eight lessons. The low-stakes, one-idea-at-a-time approach reduces cognitive overload while building a rich explanatory resource over time. Connecting today's addition to the eventual full model ('by Lesson 8 it will tell the whole story') gives students a sense of intellectual trajectory and purpose. | Collect or photograph student models before they leave, or ask students to leave them open on the desk during a brief circulation. Look for: (1) correct directional arrows showing cause and effect — not just labels floating on the page; (2) integration of Kenyan context — any student who labels their diagram with reference to sorghum storage in Turkana or bean soaking in Kisumu demonstrates high-level contextual transfer; (3) models that show dormancy and death as the same thing — flag these for a targeted warm-up correction at the start of Lesson 3. |

## D. TEACHER REFLECTION

After the lesson ask yourself: (1) Did students leave with a clear distinction between a dormant seed and a dead seed? If not, plan a one-minute clarification warm-up at the start of Lesson 3. (2) Were students able to connect the Turkana sorghum storage and Kisumu bean soaking practices to the scientific concepts of dormancy and its breaking? If these connections felt superficial, consider inviting a local farmer or agricultural extension officer to speak briefly at the start of Lesson 3, or source a short interview clip from KALRO extension media. (3) Did the DQB produce more precise questions than Lesson 1? If questions remained vague, model the transformation of a vague question into a precise one explicitly at the start of Lesson 3 before continuing. (4) Were there students who struggled to generate any question? These students may need a sentence-starter scaffold: 'I wonder if... because...' Prepare this for Lesson 3.

## E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

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| **What did I observe?** | We observed four seeds — fresh bean, soaked bean, new sorghum, and old stored sorghum — and noticed differences in seed coat texture, colour, and flexibility. We saw that soaked seeds felt softer and swollen compared to dry dormant seeds. |
| **What did I learn?** | We learned that seed dormancy has four causes: a hard impermeable seed coat, an immature embryo, chemical inhibitors such as abscisic acid, and unfavourable environmental conditions. We learned three methods of breaking dormancy: scarification, |

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| | stratification, and soaking. We connected these to practices used by farmers in Turkana and Kisumu. |
| **How does this explain the phenomenon?** | We can now explain part of the Kisumu farmer mystery: the stored seeds that did not germinate were in a state of dormancy — they were alive but their germination was blocked, possibly by a hardened seed coat or chemical inhibitors that built up during long dry storage. Dormancy is not death — it is a survival strategy that kept those seeds alive across two seasons. |

## LESSON 3 (40 minutes): What Does a Seed Need? Investigating the Conditions for Germination

### A. SPECIFIC LEARNING OUTCOMES

| Category | Specific Learning Outcomes |
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| **Purpose** | To plan and carry out a structured experiment that identifies the conditions necessary for seed germination, and to connect experimental evidence back to the Kisumu farmer phenomenon and the driving question. |
| **Knowledge** | Students will be able to state that water, oxygen, and optimum temperature are the three essential conditions for germination; explain what happens to a seed that is denied any one of these conditions; and distinguish between a control group and an experimental group. |
| **Skills** | Students will plan a fair-test experiment by identifying variables; set up experimental groups that each deny one germination condition; record daily observations in a structured evidence table; and revisit their Lesson 1 predictions to note confirmations and surprises. |
| **Attitudes** | Students will demonstrate curiosity about natural phenomena, patience and precision in recording observations, and collaborative responsibility for shared experimental set-ups. |
| **Key Inquiry Question** | Why is the concept of growth in plants important? This lesson advances the inquiry by establishing the precise chemical and physical conditions that must be met before a seed can break dormancy and begin growing, linking abstract biological concepts to observable, testable phenomena. |
| **Purpose in Storyline** | In Lesson 2 students learned WHY some seeds stay dormant. This lesson answers WHAT a seed needs to begin growing. Students move from observing existing seeds to actively manipulating conditions and gathering original evidence, which they will use in Lessons 4 through 8 to explain germination type, growth patterns, and hormone action. The experimental plants set up today will be observed again in Lesson 4. |
| **Safety Notes** | Remind students to wash hands after handling seeds and soil. Warn against putting seeds or soil near eyes or mouth. If plastic bags with boiled water are used, allow water to cool fully before sealing to avoid steam burns. Ensure all containers are labelled clearly to prevent confusion or accidental disposal. |

### B. LESSON OVERVIEW

Students open by revisiting their Lesson 1 predictions and the Kisumu farmer phenomenon, then predict what would happen to maize or bean seeds denied water, air, or warmth. They plan a four-group experiment (full conditions, no water, no air, cold temperature) using locally available materials — small transparent plastic cups or zip-lock bags, cotton wool or soil, maize or bean seeds — then set up the experiment, label groups, and begin Day 1 observation records. The lesson closes with students updating their evidence table and their Driving Question Board with new testable claims.

### C. LESSON IMPLEMENTATION FRAMEWORK

| Phase | Learner Experience | Teacher Moves | Sensemaking Strategy | Formative Assessment Strategy |
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| **Predict Phase**<br>📹 **VIDEO:** [Organism growth and the](#) | Students open their notebooks to their Lesson 1 predictions. The teacher displays or reads aloud: a quote from a student prediction | Display the Lesson 1 DQB column What we think we know. Say: Last week we learned that some seeds stay dormant because of a hard | Prediction journaling followed by partner share-out. Contrasting predictions posted on the board create productive cognitive tension | Teacher circulates and reads student prediction tables during the 60-second wait. Look for: do students identify water as essential |

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| environment<br>Source: Khan Academy (English - US curriculum)<br>🔍 Search ARES for similar videos<br><br>📖 **HTML:** Seed Dormancy and Germination - Advanced (Flexbook)<br>Source: CK-12<br>🔍 Search ARES for similar readings | written in Lesson 1 such as: the seeds need rain to grow. The teacher then poses a new prediction prompt: Imagine you are the Kisumu farmer. You have four small cups of maize seeds. In cup one you add everything the seed might need. In the other cups you remove one thing at a time. What do you think will happen in each cup? Students write individual predictions in a two-column table: Cup Setup vs What I Predict Will Happen. They then share predictions with a shoulder partner before a brief whole-class share. The teacher records three or four contrasting predictions on the board without evaluating them, leaving them visible for comparison at the end of the lesson. | coat, chemical inhibitors, or bad conditions. Today we are going to test: what exact conditions does a seed actually need to WAKE UP and start growing? Give students 2 minutes to reread their Lesson 1 predictions silently. Then say: Turn to your partner and share one prediction you made in Lesson 1 that you still believe and one you are now less sure about. Allow 2 minutes of partner talk. Call on three pairs. Then say: I am now going to give you a new scenario. The Kisumu farmer has four cups of maize seeds. Write your prediction for each cup before we plan the experiment together. WAIT TIME: After posing the cup scenario, wait a full 60 seconds of silent writing before asking anyone to share. This protects thinking time for all learners. | that motivates the investigation. | (likely yes, from prior knowledge)? Do students predict that lack of air matters (likely uncertain)? Do students predict cold will slow or stop germination (varied)? This reveals prior knowledge gaps that the experiment will address. Mentally note students who omit oxygen — these students need targeted questioning during the Observe phase. |
| **Observe Phase**<br>📹 **VIDEO:** Organism growth and the environment<br>Source: Khan Academy (English - US curriculum)<br>🔍 Search ARES for similar videos<br><br>📖 **HTML:** Seed Dormancy and Germination - Advanced (Flexbook)<br>Source: CK-12<br>🔍 Search ARES for similar readings | Students work in groups of four to set up the experiment. Each group receives: 16 maize or bean seeds of roughly equal size, four transparent plastic cups or zip-lock bags, cotton wool or a small amount of garden soil, water, a permanent marker for labelling, and a student experiment planning sheet. Groups first complete the planning sheet by identifying: the independent variable (the condition being changed), the dependent variable (whether germination occurs), and the controlled variables (same seed type, same amount of cotton wool or soil, same light exposure). They then set up four cups labelled as follows. Cup A - Control: seeds on moist cotton wool, open to air, kept at room temperature (approximately 20 to 30 degrees Celsius in a Kenyan classroom). | Distribute materials and the planning sheet. Say: Before you touch any seeds, your group must agree on what you are testing, what you are keeping the same, and how you will know if the seed has germinated. Give groups 4 minutes to complete the planning sheet. Then say: Raise your hand if your group agrees that Cup A is the control. Why is it the control? Call on one group to explain. Clarify if needed: the control cup has ALL the conditions we think a seed needs. Every other cup removes exactly ONE condition so we can see what difference that one condition makes. This is called a fair test. During set-up, circulate and use these targeted prompts: To a group unsure about Cup C say - How can you make sure the seeds in Cup C have very little air available to them? Think about | Structured planning sheet forces explicit identification of variables before hands-on work begins, building scientific reasoning habits. The baseline Day 1 observation anchors the evidence table in real data rather than memory. | Collect or photograph one planning sheet per group before they begin set-up. Check: have students correctly identified the independent variable? Have they listed at least three controlled variables? Are their Day 1 observations descriptive (colour, size, texture) rather than vague? Groups with incomplete variable identification should be redirected before they begin set-up to prevent flawed experimental design. |

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| | Cup B - No Water: seeds on dry cotton wool, open to air, room temperature. Cup C - No Air: seeds on moist cotton wool submerged under a layer of water in a sealed cup to limit air contact. Cup D - Cold Temperature: seeds on moist cotton wool, open to air, placed in a cool location such as a water bucket or the coolest corner of the school, or wrapped in a damp cloth and placed in a shaded area. Students record Day 1 observations immediately: seed appearance, colour, size, any visible change. The teacher circulates, asking probing questions rather than giving answers. | what happens when something is fully covered by water. To a group about to skip the planning sheet say - If you do not write your plan first, how will you know tomorrow whether you followed the same procedure for every cup? To an advanced group say - What do you predict the seed will look like after three days in each cup? Can you add a prediction sketch to your evidence table? After all groups have set up their cups, say: Now make your Day 1 observation. Record what the seeds look like RIGHT NOW before anything changes. This is your baseline. WAIT TIME: After asking Why is Cup A the control? wait 30 seconds before calling on a student. After a student answers, ask a follow-up: Does anyone want to add to or challenge that explanation? Wait another 20 seconds. | | |
| **Explain Phase**<br>📹 **VIDEO:**<br>Organism growth and the environment<br>Source: Khan Academy (English - US curriculum)<br>🔍 Search ARES for similar videos<br><br>📖 **HTML:**<br>Seed Dormancy and Germination - Advanced (Flexbook)<br>Source: CK-12<br>🔍 Search ARES for similar readings | After set-up, groups pause to discuss and write a brief scientific explanation connecting today's experiment to prior knowledge from Lessons 1 and 2. The teacher poses three explain prompts on the board: (1) Based on what you already know, explain WHY you predict Cup B seeds will not germinate. Use the word imbibition or water uptake in your answer. (2) We learned in Lesson 2 that abscisic acid keeps seeds dormant. Which cup condition might be similar to how abscisic acid works, and why? (3) The Kisumu farmer stored bean seeds for two seasons in a dry tin. Using today's experiment, explain why those seeds did not germinate in the tin. Students write responses individually for 4 minutes, then | After groups have set up their cups and completed Day 1 observations, say: Now put your pencils down for a moment. You have set up an experiment but science does not stop at doing the experiment. You also need to explain the thinking behind it. I am going to put three explain prompts on the board. You have 4 minutes to write your own answers before you talk to your group. Display the three prompts. WAIT TIME: Enforce the 4 minutes of silent individual writing strictly. Say: I know some of you want to talk to your group immediately. Please do not. Your brain needs to try the explanation first before you hear someone else is idea. After 4 minutes, say: Now compare your explanations with your group. Do | Individual writing before group discussion protects epistemic agency. Using student disagreements as data points models authentic scientific discourse. The explicit connection back to the Kisumu farmer keeps the phenomenon central. | Circulate during the 4 minutes of individual writing. Specifically check: does the student connect water to enzyme activation or cell expansion (imbibition)? Does the student connect oxygen to respiration and energy release? Does the student connect the dry tin storage to the No Water condition in their experiment? Students who write only one-word answers (seeds need water to grow) need a prompt such as: Can you explain what water actually does inside the seed after it soaks in? This reveals depth of understanding beyond recall. |

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| | compare answers within the group. Groups share one insight with the class. The teacher uses student responses to build a class explanation on the board connecting the three conditions (water, oxygen, temperature) to the cellular processes that start germination: water triggers imbibition which activates enzymes; oxygen supports aerobic respiration to release energy for cell division; optimum temperature ensures enzyme activity is at its peak. | you all agree? If not, discuss the disagreement. After 3 minutes of group discussion, ask: Which group found a disagreement inside their group? What were the two different ideas? Use disagreements as teachable moments. Avoid resolving disagreements by simply giving the answer. Instead say: What evidence from today is experiment or from Lesson 2 would help you decide which explanation is better? Close the Explain phase by building the three-condition summary on the board using student language as much as possible. Connect explicitly to the phenomenon: So back to our Kisumu farmer. She waters her seeds. She plants them in open soil. The temperature is warm. According to your experiment design, what should happen? Why might some seeds still fail? | | |
| **Driving Question Board (DQB) Creation**<br>📹 **VIDEO:**<br>Organism growth and the environment<br>Source: Khan Academy (English - US curriculum)<br>🔍 Search ARES for similar videos<br><br>📖 **HTML:**<br>Seed Dormancy and Germination - Advanced (Flexbook)<br>Source: CK-12<br>🔍 Search ARES for similar readings | Students return to the class DQB posted on the wall from Lessons 1 and 2. The teacher facilitates a 5-minute whole-class update. Students are invited to: (a) tick or star any question from the DQB that today is experiment has begun to answer; (b) add at least two new questions inspired by the experiment set-up. Likely new questions include: What happens inside the seed when water enters? Does more water always mean faster germination? Why does cold stop germination but freezing sometimes helps (stratification)? How does the seed know when the temperature is right? Students write their personal new questions on sticky notes and add them to the DQB under a new column labelled New Questions | Direct students to the DQB. Say: In Lesson 1 we wrote what we think we know and what we wonder. In Lesson 2 we added questions about dormancy. Today you have run your own experiment. Look at the questions already on the board. Which ones can you now say we have started to answer? Give students 90 seconds to read the board silently and put a small tick on any question they feel the experiment begins to address. Then say: Now think about what this experiment made you MORE curious about. Not less curious — more. Write two new questions on your sticky notes. You have 2 minutes. WAIT TIME: After saying write two new questions, wait the full 2 minutes without prompting. Silence is | The DQB functions as a living knowledge artefact. Returning to it each lesson makes the cumulative nature of scientific inquiry visible. Ticking answered questions provides metacognitive feedback and motivates continued inquiry. | Read the new sticky-note questions as students post them. Questions that reveal misconceptions (for example: does the seed eat the water?) should be highlighted and gently reframed rather than dismissed. The quality and specificity of new questions is itself a formative indicator: vague questions suggest surface engagement while mechanistic questions (what enzyme breaks down the seed coat?) suggest deeper processing. |

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| | from Lesson 3 Experiment. The teacher points out the connection between the DQB and the storyline ahead: Lesson 4 will show what happens AFTER germination begins, and the cups from today will be their evidence. | productive here. After sticky notes are posted, read three or four aloud and say: Notice that good experiments do not just answer questions. They create new ones. That is how science works. Point to the DQB arc and say: By Lesson 8 we want to be able to answer ALL of these questions. Hold onto your curiosity. | | |
| **Model Building Phase**<br>📹 **VIDEO:** [Organism growth and the environment]<br>Source: Khan Academy (English - US curriculum)<br>🔍 [Search ARES for similar videos]<br><br>📖 **HTML:** [Seed Dormancy and Germination - Advanced (Flexbook)]<br>Source: CK-12<br>🔍 [Search ARES for similar readings] | In the final 5 minutes, students individually draw a simple initial model in their notebooks. The model prompt is: Draw and annotate a diagram showing what must happen for a seed to move from dormancy to germination. Your diagram must include: the three conditions (water, oxygen, optimum temperature); an arrow showing the direction of change from dormant seed to germinating seed; and at least one connection to what you learned in Lesson 2 about breaking dormancy. Students are told this is an initial model — it does not need to be perfect or complete. They will return to it and revise it in Lessons 4, 5, 6, and 8. The teacher emphasises: a model is not a drawing of what you see. It is a diagram that explains WHY and HOW something happens. Students label their diagram with today is date and Lesson 3 Initial Model. | Signal the final phase by saying: We have observed, we have planned, we have explained. Now I want you to build a model. A model in science is not just an illustration. It is an explanation in diagram form. Open your notebook to a fresh page. You have 5 minutes to draw your Lesson 3 initial model. Display the three model requirements on the board. Circulate and use targeted prompts: To a student drawing only a seed say: Your diagram shows what the seed looks like. Now add arrows or labels to show what the three conditions DO to the seed. To a student who draws elaborate detail say: Which part of your diagram explains HOW water triggers germination? Can you add a label that explains the process not just names it? After 5 minutes say: Place your pencil down. Do not worry if it feels incomplete. Science models are always works in progress. You will revise this model five more times before Lesson 8. The incompleteness is the point — it shows you what you still need to learn. Close by saying: Tomorrow, check your cups at home or we will check them together. Your experiment is running right now. The seeds are deciding whether to wake up. Come back next lesson ready to | Initial model construction forces integration of procedural knowledge (the experiment) with conceptual knowledge (why conditions matter). Framing it as a work in progress reduces anxiety and builds a revision habit that cumulates through the storyline. | Quick gallery walk: teacher views 6 to 8 models in 3 minutes during the drawing phase. Look for: does the model show directionality from dormancy to germination? Does it include at least two of the three conditions? Does it make any mechanistic claim (water enters seed and activates enzymes) or only label conditions? Models with no mechanistic language flag students who need a more explicit bridge between condition and biological process in Lesson 4. |

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| | | observe what they decided. | | |

| D. TEACHER REFLECTION |
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| After the lesson, consider: Did all groups successfully identify independent and controlled variables before set-up, or did some groups need re-direction? Were there any students who could not connect today is experiment to the Lesson 2 dormancy content? If so, these students may need a brief revision moment at the start of Lesson 4 before the germination-type observation begins. Check that all experimental cups are stored safely and consistently — inconsistent storage over 48 to 72 hours will compromise Day 3 observations used in Lesson 4. Consider taking a photograph of each group is cup set-up as a reference record. Also reflect: did students ask mechanistic questions on the DQB (about enzyme activation, energy release, cell division) or only descriptive questions? Mechanistic questioning signals readiness for the hormone content in Lesson 6. If most questions were descriptive, plan to scaffold more deeply during the Lesson 5 cambium activity before introducing hormones. |

| E. SUMMARY TABLE PROMPT (pre-filled example for this lesson) | |
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| **What did I observe?** | We observed that maize and bean seeds placed on moist cotton wool in transparent cups showed no visible change on Day 1. We set up four experimental groups: one with full conditions, one without water, one without air, and one in a cold location. We recorded our baseline observations and noted that on Day 1 all seeds looked similar regardless of the condition they were placed in. |
| **What did I learn?** | We learned that three conditions are essential for germination: water (which triggers imbibition and enzyme activation), oxygen (which powers aerobic respiration to release energy for cell growth), and optimum temperature (which ensures enzyme activity is efficient). We connected these conditions to the Kisumu farmer is garden: her seeds had water, air, and warmth, which is why most germinated. We also connected the No Water cup to the dry tin storage from Lesson 2 — seeds in dry conditions cannot begin germination regardless of other factors. |
| **How does this explain the phenomenon?** | We explained that the Kisumu farmer is stored seeds that failed to germinate were likely in conditions similar to our Cup B — no water available. We also explained that a seed does not just need to have dormancy broken (Lesson 2) — it then actively needs three external conditions to be met simultaneously before germination can begin. Our experiment models this: removing even one condition is enough to prevent or delay germination, which is why the farmer is seeds respond so differently even when planted in the same garden on the same day. |

| LESSON 4 (40 minutes): Two Ways to Emerge: Epigeal and Hypogeal Germination | |
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| **A. SPECIFIC LEARNING OUTCOMES** | |
| **Category** | **Specific Learning Outcomes** |
| **Purpose** | Students will observe, compare, and explain the two types of seed germination using real germinating maize and bean seeds in transparent containers, connecting their observations to the anchoring phenomenon of the Kisumu farmer who was puzzled by the striking difference in how her bean and maize seedlings emerged from the soil. |
| **Knowledge** | Students will be able to: (1) define epigeal germination as the type in which the cotyledons are pushed above the soil surface by the elongating hypocotyl; (2) define hypogeal germination as the type in which the cotyledons remain below the soil surface while the epicotyl pushes the plumule upward; (3) identify the structures involved in each type (hypocotyl, epicotyl, cotyledon, plumule, radicle); (4) give locally relevant examples: beans and groundnuts for epigeal, maize and wheat for hypogeal. |
| **Skills** | Students will practise scientific observation by examining germinating seeds in transparent containers, produce accurate labelled diagrams of epigeal and hypogeal germination, compare and contrast the two types using a structured Venn diagram, and construct written explanations linking structure to function. |
| **Attitudes** | Students will develop curiosity and appreciation for the diversity of plant strategies, demonstrate precision and care when handling fragile seedlings, and build confidence in using scientific vocabulary to explain natural phenomena they have observed in everyday Kenyan agricultural settings. |
| **Key Inquiry Question** | Why is the concept of growth in plants important? In this lesson students answer this inquiry by discovering that even the very first act of growth after germination — how the seedling emerges — follows a genetically controlled pattern that determines where stored food is used and how the seedling survives its first days, which has direct consequences for farmers managing crop emergence. |
| **Purpose in Storyline** | Lesson 4 is the Observe and Explain turning point in the evidence-gathering sequence. Students now have the evidence from Lesson 3 (water, oxygen, temperature trigger germination) and they return to the original puzzle: why do maize and bean seedlings look so different when they emerge? This lesson resolves the core visual mystery of the anchoring phenomenon by giving students a mechanistic explanation. The Kisumu farmer observation is now fully explained for the first time, and students update the Driving Question Board with new understanding before moving to Lesson 5 where they explore how plants continue to grow in length and girth. |
| **Safety Notes** | Handle germinating seedlings gently to avoid breaking delicate radicles and plumules. Wash hands after handling soil or moist cotton wool. If transparent containers have been sealed, open carefully to avoid spills. No chemical hazards in this lesson. Students with skin sensitivities should avoid prolonged contact with moist soil; paper towels are an alternative germination medium. |

## B. LESSON OVERVIEW

Students begin by revisiting their Lesson 1 predictions about why bean and maize seedlings emerge differently. They examine live germinating bean and maize seeds in transparent containers prepared since Lesson 3 and record detailed observations. Through guided comparison they discover that bean seedlings pull their cotyledons above the soil (epigeal) while maize cotyledons stay underground (hypogeal). Students draw labelled diagrams, complete a Venn diagram comparison, and write brief explanations. The lesson closes with a DQB update and a simple visual model sketch showing both germination pathways side by side.

## C. LESSON IMPLEMENTATION FRAMEWORK

| Phase | Learner Experience | Teacher Moves | Sensemaking Strategy | Formative Assessment Strategy |
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| **Predict Phase**<br>📹 **VIDEO:**<br>Organism growth and the environment<br>Source: Khan Academy (English - US curriculum)<br>🔍 Search ARES for similar videos<br><br>📖 **HTML:**<br>Seed Dormancy and Germination - Advanced (Flexbook)<br>Source: CK-12<br>🔍 Search ARES for similar readings | Students open their notebooks to their Lesson 1 prediction entry (approximately 3 minutes). The teacher displays or reads aloud the original anchoring phenomenon description: the Kisumu farmer notices bean seedlings forming an arch above the soil while maize sends up a single green spike with the seed staying underground. Students re-read their original predictions silently (1 minute), then turn to a partner and answer: What did I think was happening then, and do I still think the same thing now after Lessons 2 and 3? Partners share briefly. The teacher then poses a focused prediction question on the board: Which part of the bean seedling do you think pushes the cotyledons above the ground? Which part of the maize seedling pushes the shoot upward while the seed stays buried? Students write a two-sentence prediction in their evidence table before examining the specimens. | Teacher displays the anchoring phenomenon photograph (or re-shows the transparent containers from Lesson 1) alongside a simple sketch on the board showing a bean seedling arch above soil and a maize spike above soil with a buried seed. Teacher says: We spent Lesson 3 finding out what a seed needs to begin germinating. Today we are going to look very carefully at what happens next — the actual emergence of the seedling — and I think you are going to be able to explain the Kisumu farmer puzzle by the end of this lesson. Before we look at our specimens, I want you to go back to your very first prediction from Lesson 1. Read it. Has anything you learned in Lessons 2 or 3 changed your thinking? Take 60 seconds to read silently. [WAIT 60 seconds — teacher scans room to check engagement, does not speak.] Teacher then asks: Turn to your partner. Tell them one thing your prediction got right and one thing you are now unsure about. [WAIT 90 seconds for partner talk.] Teacher cold-calls two pairs using name sticks, affirming both confident and uncertain responses: It is completely fine to say you are still unsure — that is exactly why we observe. Teacher writes the prediction question on the board and gives students 2 minutes to write their two-sentence prediction. [WAIT 2 minutes — teacher circulates, reads over shoulders, notes 2 or 3 interesting predictions to reference later without naming students.] | Activating prior knowledge through prediction revision — students compare their current thinking to their Lesson 1 thinking, making their conceptual change visible. Partner talk lowers the risk of being wrong publicly and ensures all students engage with the question before observation begins. | Teacher scans written predictions during circulation. Key diagnostic indicators: (1) Do students mention any specific structure (hypocotyl, stem, shoot) or do they use vague language like just the plant pushes up? (2) Do any students already use the words epigeal or hypogeal from prior knowledge or reading? (3) Are students connecting to the seed energy / cotyledon food-store idea from Lesson 2? Teacher notes gaps to address in the Explain phase. |
| **Observe** | Groups of 4 students receive one | Teacher distributes containers | Structured silent observation | Teacher listens for: (1) accurate |

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| **Phase**<br>📹 **VIDEO:**<br>[Organism growth and the environment]<br>Source: Khan Academy (English - US curriculum)<br>🔍 Search ARES for similar videos<br><br>📖 **HTML:**<br>[Seed Dormancy and Germination - Advanced (Flexbook)]<br>Source: CK-12<br>🔍 Search ARES for similar readings | transparent container with germinating bean seeds (approximately 5 to 7 days old) and one transparent container with germinating maize seeds (approximately 5 to 7 days old), prepared since Lesson 3. Each container is placed against the transparent side so root and shoot development is visible. Students also receive a printed or hand-drawn observation sheet with two outlines: one blank seedling space labelled BEAN and one labelled MAIZE. Step 1 (4 minutes): Students observe the bean container silently and individually, writing or sketching what they see. Prompt on board: Where are the cotyledons? What shape is the seedling making above the soil? What is below the soil? Step 2 (4 minutes): Students observe the maize container and repeat the process. Prompt on board: Where is the grain (seed)? Where is the green shoot emerging from? What is happening underground? Step 3 (3 minutes): Students share observations within their group and agree on the most important differences to report. The teacher pauses the class and asks each group to state one observation using the sentence frame: In the bean seedling we noticed BLANK, but in the maize seedling we noticed BLANK. | carefully and says: You have about 10 minutes to observe both specimens very carefully. Do not pull the seedlings out — we want to keep them alive and intact. Use your observation sheet to sketch what you see from the outside of the container. Start with the bean container. I want absolute silence for the first 4 minutes so everyone can look carefully before talking. [WAIT 4 minutes — teacher circulates, crouching to group level, pointing gently at features without naming them, e.g., Look at where the thick curved part is. Where does that part start? Where does it end?] After 4 minutes teacher says quietly: Now move to the maize container. Same careful observation. Another 4 minutes of quiet sketching. [WAIT 4 minutes — teacher continues circulating, asking probing questions to groups that finish early: Can you see where the green shoot is coming from inside the maize grain? What is still inside the soil?] After maize observation teacher says: Share with your group. You have 3 minutes. Then each group will give me one difference using the sentence frame on the board. [WAIT 3 minutes.] Teacher calls on each group in turn, recording key observations on the board in two columns headed BEAN and MAIZE. Teacher resists labelling or explaining at this stage — the goal is to capture raw observations in student language. | before discussion ensures all students form their own observations before being influenced by peers. The sentence frame (In the bean we noticed... but in the maize we noticed...) scaffolds comparative language and keeps observations precise. Teacher recording student observations on the board validates student voice and creates a shared data set for the Explain phase. | location of cotyledons in each specimen (above soil for bean, underground for maize); (2) use of shape language (arch, hook, curved for bean; straight spike for maize); (3) any student who spontaneously uses the words epigeal or hypogeal; (4) misconceptions such as the maize seed dissolves underground or the bean arch is caused by wind. Teacher notes misconceptions to address directly in the Explain phase with a targeted question rather than a correction. |
| **Explain Phase**<br>📹 **VIDEO:**<br>[Organism growth and the environment]<br>Source: Khan | Teacher introduces the two terms epigeal and hypogeal using the board observations as the starting point. Students label their sketches using a guided labelling activity. Each student then completes a | Teacher begins the explain phase by pointing to the board observations and saying: You have just described two completely different strategies for getting a seedling out of the soil. Scientists | Concept introduction is anchored in student-generated observations rather than a definition-first approach, ensuring meaning is constructed from evidence. The Venn diagram externalises | Teacher reads 6 to 8 Venn diagrams during circulation to check for: (1) correct placement of cotyledon position in the exclusive zones; (2) inclusion of structure names (hypocotyl, epicotyl) not |

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| Academy (English - US curriculum)<br>🔍 Search ARES for similar videos<br><br>📖 **HTML:** Seed Dormancy and Germination - Advanced (Flexbook)<br>Source: CK-12<br>🔍 Search ARES for similar readings | printed or copied Venn diagram in their notebook with the circles labelled EPIGEAL GERMINATION (BEANS) and HYPOGEAL GERMINATION (MAIZE) and the overlap labelled BOTH. Suggested content: epigeal only — cotyledons above soil, hypocotyl elongates and arches, cotyledons act as first photosynthetic leaves; hypogeal only — cotyledons remain underground, epicotyl elongates and pushes plumule up, cotyledons act purely as food store; both — radicle emerges first, seed coat splits, needs water and oxygen, has a plumule, roots anchor the seedling. After completing the Venn diagram, students write a 3-sentence explanation in their evidence table answering: Why does the Kisumu farmer see such different shapes when her bean and maize seeds germinate? Students share one sentence each with the class and the teacher compiles a class explanation on the board. Students add any missing points to their own explanation. | have given these two strategies names. Listen carefully. In epigeal germination — epi means above in Greek — the cotyledons come above the soil. In hypogeal germination — hypo means below — the cotyledons stay below. Now look at your bean sketch. Which type is it? [WAIT 10 seconds.] And the maize? [WAIT 10 seconds.] Good. Now let us understand WHY. Teacher draws a simple annotated diagram on the board for each type, labelling: radicle, hypocotyl (bean — elongates, forms the arch), cotyledons (bean — above soil; maize — below soil), epicotyl (maize — elongates to push plumule up), plumule. Teacher says: I want you to label your own sketches using these labels. Take 3 minutes. [WAIT 3 minutes.] Teacher then introduces the Venn diagram task and says: I want you to sort what you know into the Venn diagram. Work individually for 4 minutes, then you can check with your partner. [WAIT 4 minutes, then 2 minutes partner check.] Teacher then asks: Now write three sentences that explain the Kisumu farmer puzzle. Start your first sentence with the word Beans... and your second with Maize... and your third with This means the farmer sees... [WAIT 3 minutes for writing.] Teacher cold-calls 3 students to share one sentence each, builds the class explanation on the board, and asks: Does anyone have a sentence that says something not yet on the board? Add it. Teacher closes the explain phase by connecting to cotyledon function from Lesson 2: Remember in Lesson 2 we said cotyledons store food. Does it matter whether the | comparative thinking. The sentence frame (Beans... Maize... This means the farmer sees...) scaffolds explanation writing while keeping the anchoring phenomenon central. Connecting cotyledon position to function (photosynthesis vs. pure food storage) deepens the mechanistic explanation beyond naming. | just vague descriptions; (3) any student who confuses the hypocotyl and epicotyl roles. Three-sentence explanations are collected at the end of the lesson as a written formative product. Teacher scans for whether students connect the explanation back to the farmer context (not just definitions in isolation). |

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| | | food store is above or below the soil? What are the risks for each strategy? [WAIT 30 seconds — partner talk, then two responses collected.] | | |
| **Driving Question Board (DQB) Creation**<br>📹 **VIDEO:** Organism growth and the environment<br>Source: Khan Academy (English - US curriculum)<br>🔍 Search ARES for similar videos<br><br>📖 **HTML:** Seed Dormancy and Germination - Advanced (Flexbook)<br>Source: CK-12<br>🔍 Search ARES for similar readings | The teacher directs students to the class DQB (physical sticky-note board or a section of the chalkboard divided into What we think we know, What we wonder, and What we have now explained). Students individually write one sticky note or board entry (2 minutes): either something they now understand that was previously in the What we wonder column, or a new question that today's lesson has raised. Students place or read their entries. As a class, the group moves at least two questions from What we wonder to What we have now explained: specifically the question about why bean and maize seedlings look different when they emerge, and the question about where the seed goes after germination in maize. Students also check whether the Kisumu farmer puzzle is now answered. | Teacher says: Look at our DQB. When we started this unit, we had questions we could not answer. Let us check. Read through the What we wonder column silently for 30 seconds. [WAIT 30 seconds.] Now I want each of you to write one sticky note: either move a question to explained, or add a brand new question today raised for you. You have 2 minutes. [WAIT 2 minutes.] Teacher invites 4 to 5 students to share and place their notes. Teacher guides the class to formally move the bean vs. maize emergence question to the explained column and writes a one-sentence class explanation beside it: Bean seeds use epigeal germination with the hypocotyl pulling cotyledons above soil, while maize uses hypogeal germination with the epicotyl pushing the plumule above soil while cotyledons stay underground. Teacher then says: What new questions does today raise? Is there anything about how the plant keeps growing after it emerges that we have not yet explained? [WAIT 20 seconds — this seeds curiosity for Lesson 5 on primary and secondary growth.] | The DQB update ritualises evidence accumulation and makes conceptual progress visible to students. Moving questions from wonder to explained gives students a sense of epistemic accomplishment. Seeding new questions for Lesson 5 maintains narrative momentum and intellectual curiosity. | Teacher reviews the new questions students raise on the DQB. Key indicators of readiness for Lesson 5: students asking about how the plant gets taller, why some plants get thick trunks and others stay thin, or what controls which direction the plant grows. These questions indicate students are ready for primary and secondary growth. If most new questions are still about germination conditions, the teacher knows to briefly consolidate before moving forward. |
| **Model Building Phase**<br>📹 **VIDEO:** Organism growth and the environment | In the final 5 minutes, each student draws a side-by-side sketch model in their notebook titled Two Ways to Emerge. The model shows a cross-section of soil with a bean seedling on the left and a maize seedling on the right, each at | Teacher says: I want you to finish today by building a small model in your notebook — not a perfect diagram, but a thinking sketch. Show me both types of germination side by side. Label the key parts and draw an arrow | The side-by-side sketch model externalises student understanding in a spatial format that directly mirrors the anchoring phenomenon visual. Requiring directional arrows forces students to think mechanistically (what | Teacher collects or photographs 6 to 8 models during circulation. Evaluation criteria: (1) correct cotyledon position in each type; (2) correct structure labelled as responsible for elongation (hypocotyl vs. epicotyl); (3) |

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| Source: Khan Academy (English - US curriculum)<br>🔍 Search ARES for similar videos<br><br>📖 **HTML:** Seed Dormancy and Germination - Advanced (Flexbook)<br>Source: CK-12<br>🔍 Search ARES for similar readings | approximately day 5 of germination. Students must label at minimum: soil surface line, radicle, cotyledon (with its position relative to soil), hypocotyl (bean) or epicotyl (maize), and plumule. Students add one annotation arrow for each seedling explaining the direction of force — what is pushing or pulling what. Below the sketch students write one sentence: This model explains the Kisumu farmer puzzle because BLANK. | showing what is doing the work of pushing or pulling the seedling out of the soil. You have 4 minutes. [WAIT 4 minutes — teacher circulates and selects 2 models that show strong labelling and 1 that has a common error such as placing the cotyledon below soil for the bean.] Teacher then holds up (or describes without naming the student) the strong models and says: This model clearly shows the hypocotyl arching above the soil in the bean. Notice how the arrow shows the direction of elongation. This is exactly what the Kisumu farmer saw. Teacher addresses the error model by asking: If the cotyledon is below the soil in this bean diagram, what would the farmer see above ground? Would she see the arch shape? [WAIT 15 seconds.] Students self-correct. Teacher closes by saying: Next lesson we will ask: after the seedling is out — how does it keep growing taller, and why do some plants eventually grow thick and woody while others stay thin? Write that question at the bottom of your model page as your exit curiosity. | structure is doing the work) rather than just naming parts. The closing sentence anchors the model to the real-world context. The exit curiosity question is a low-stakes preview that activates anticipation for Lesson 5. | direction of elongation arrows correct; (4) closing sentence connects back to the farmer context. Models with the cotyledon placed incorrectly or arrows pointing the wrong way are flagged for a brief 2-minute clarification at the start of Lesson 5 before new content begins. |

## D. TEACHER REFLECTION

After the lesson consider the following questions: (1) Were students able to transfer the term cotyledon from the anchoring phenomenon discussion in Lesson 1 to today's labelling activity, or did many students seem to encounter it as a new word? If so, plan a 2-minute vocabulary consolidation at the start of Lesson 5. (2) Did the transparent containers show sufficient germination to make the difference between epigeal and hypogeal clearly visible? If bean seedlings had not yet arched above the soil, the observation phase may have been limited — consider extending germination time to day 6 or 7 before Lesson 4 in future iterations. (3) Were students able to connect cotyledon position to function (photosynthesis above ground vs. pure food storage below ground) or did they focus only on position without considering function? If function was missed, build a bridging question into the start of Lesson 5: What happened to the cotyledons in the bean plant after the first true leaves opened? (4) Did the DQB update generate genuine new questions about continued growth, or did students simply paraphrase what they had just learned? Genuine new questions (about trunks, height, thickness) signal readiness to advance. (5) Were any students distracted by the living specimens and unable to focus on structured observation? Consider assigning specific roles (observer, sketcher, reporter, timer) in future group work to maintain focus.

| E. SUMMARY TABLE PROMPT (pre-filled example for this lesson) | |
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| **What did I observe?** | Students observed germinating bean seeds with curved arch-shaped seedlings pushing yellow-green cotyledons above the soil surface, and germinating maize seeds with a straight green spike emerging above the soil while the grain remained buried underground. They noted that the bean seedling forms a distinct hook or arch shape while the maize seedling emerges as a straight upright point. |
| **What did I learn?** | Students learned that the two types of seed germination are called epigeal (cotyledons emerge above the soil, driven by hypocotyl elongation — as in beans and groundnuts) and hypogeal (cotyledons remain below the soil while the epicotyl pushes the plumule upward — as in maize and wheat). They labelled diagrams with radicle, hypocotyl, epicotyl, cotyledon, and plumule, and completed a Venn diagram comparing both types. |
| **How does this explain the phenomenon?** | Students explained the Kisumu farmer puzzle: the bean seedling forms an arch because the hypocotyl elongates and pulls the cotyledons above the soil where they can begin photosynthesis, while the maize seedling sends only its plumule upward via the elongating epicotyl, keeping the energy-rich cotyledon safely underground as a food reserve. Both strategies successfully get the seedling established, but through different structural mechanisms determined by the genetics of each plant species. |

| LESSON 5 (40 minutes): Getting Taller vs. Getting Thicker: Primary and Secondary Growth | |
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| **A. SPECIFIC LEARNING OUTCOMES** | |
| **Category** | **Specific Learning Outcomes** |
| **Purpose** | Students will distinguish between primary growth and secondary growth in plants, connecting each growth type to specific meristematic tissues and explaining how they together account for the differences between herbaceous plants and woody trees observed in the anchoring phenomenon. |
| **Knowledge** | Students will be able to: (1) define primary growth as an increase in length driven by apical meristems at root and shoot tips; (2) define secondary growth as an increase in girth driven by lateral meristems, specifically the vascular cambium; (3) identify the vascular cambium in a cross-section of a woody stem; (4) explain why herbaceous plants such as sukuma wiki show little or no secondary growth while woody plants such as mango and avocado show extensive secondary growth; (5) connect annual ring formation in woody stems to seasonal growth patterns in Kenyan trees. |
| **Skills** | Observing and comparing plant stem cross-sections; drawing and labelling scientific diagrams; constructing an initial explanatory model; using evidence from specimens to support a claim; collaborating in small groups to analyse and present observations. |
| **Attitudes** | Curiosity about the invisible cellular processes that produce visible plant structures; appreciation for how scientific explanations connect observable phenomena to underlying mechanisms; confidence in using evidence to revise prior ideas. |
| **Key Inquiry Question** | Why is the concept of growth in plants important? This lesson answers the inquiry by showing that two distinct growth processes — primary and secondary — explain the differences between thin herbaceous crops and massive woody trees, both of which are vital to Kenyan agriculture and food security. |
| **Purpose in Storyline** | Lessons 1 to 4 explained how seeds germinate and emerge from the soil in two different ways. This lesson advances the storyline by asking: what happens after a seedling emerges? It introduces the two mechanisms that explain why the bean plants in the Kisumu farmer garden stay relatively short and soft while the nearby mango and avocado trees grow taller and develop thick woody trunks year after year. This sets up Lesson 6, which will explain the hormonal signals that regulate these growth processes. |
| **Safety Notes** | Handle fresh stem cross-sections with care; if razor blades or scalpels are used by the teacher to prepare sections, students must not handle cutting instruments — pre-cut sections should be prepared before class. Wash hands after handling plant material. If a mango twig is sourced from outside, check for latex sap that may irritate sensitive skin. |

## B. LESSON OVERVIEW

In this 40-minute Explain Phase lesson, students first predict what they think happens inside a plant stem as a tree grows thicker over many years. They then observe and compare prepared cross-sections of a herbaceous stem (sukuma wiki or bean) with a woody stem (mango twig or avocado branch), guided by a labelled reference diagram. Through structured sense-making, students explain the roles of apical meristems in primary growth and vascular cambium in secondary growth. They build an initial annotated model of a woody stem cross-section, update the Driving Question Board, and connect their explanation to the Kisumu farmer anchoring phenomenon — specifically, why the mango and avocado trees next to the bean plants grew thicker trunks over years while the bean plants did not.

## C. LESSON IMPLEMENTATION FRAMEWORK

| Phase | Learner Experience | Teacher Moves | Sensemaking Strategy | Formative Assessment Strategy |
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| **Predict Phase**<br>📹 **VIDEO:**<br>Sinan, Rüstem Pasha Mosque<br>Source: Khan Academy (English - US curriculum)<br>🔍 Search ARES for similar videos<br><br>📖 **PDF:**<br>Orange-book-primary-2017<br>Source: Kenya Curriculum Tools 2025<br>🔍 Search ARES for similar readings | Students are shown two photographs side by side on the board or a printed handout: (1) a young sukuma wiki stem, thin and green, snapped cleanly; (2) a cross-section of a mango tree trunk showing visible rings and thick brown bark. The teacher asks students to write individually for 2 minutes in their notebooks: 'What do you think is happening INSIDE these two stems that makes them so different? Where do you think new material comes from to make the mango trunk thicker each year?' Students share predictions with a shoulder partner before a few are shared with the class. Predictions are recorded on a visible 'Our Predictions' section of the board without correction, so they can be revisited at the end of the lesson. | Display the two photographs clearly and give students a moment of silent looking before writing. Say: 'Before we look at any evidence, I want you to commit to an idea. Write your best thinking — even if you are not sure. A wrong prediction that we revise is still great science.' Allow 2 minutes of silent individual writing. Then say: 'Share your prediction with your partner. You have 1 minute.' [WAIT TIME: 1 minute partner talk.] Cold-call three to four pairs using a name stick or random selection. Record key words from each prediction on the board — do not affirm or correct. Say: 'We are going to keep these predictions visible. By the end of the lesson, you will decide which ones the evidence supports.' Do NOT introduce vocabulary yet — allow predictions to use everyday language. | Prediction anchoring — students commit to an idea before observing, which creates cognitive tension that drives engagement during the Observe Phase. Recording predictions publicly creates accountability and a reference point for revision. | Listen for whether students connect the mango trunk thickness to something happening inside the stem versus only surface-level explanations such as 'it just grows.' Identify students who already intuit that cells must be dividing somewhere in the middle of the stem — these students can be called on during the Explain Phase to share their thinking. Note students who predict that the whole stem grows thicker uniformly — this is a key misconception to address. |
| **Observe Phase**<br>📹 **VIDEO:**<br>Sinan, Rüstem Pasha Mosque<br>Source: Khan Academy (English - US curriculum)<br>🔍 Search ARES for similar videos<br><br>📖 **PDF:**<br>Orange-book-primary-2017<br>Source: Kenya Curriculum Tools 2025<br>🔍 Search ARES for similar readings | Students work in groups of four. Each group receives: (a) a pre-cut fresh cross-section of a sukuma wiki stem (or bean stem) placed on a piece of white paper; (b) a pre-cut fresh cross-section of a mango twig or avocado branch of approximately 1 to 2 cm diameter, also on white paper; (c) a printed or teacher-drawn reference diagram showing the labelled parts of both a herbaceous and a woody stem cross-section, including epidermis, cortex, vascular bundles (herbaceous), vascular cambium (woody), xylem, phloem, pith, and bark. Students use a hand lens if available to examine both specimens. They complete an observation table in their | Circulate to each group during observation. Do not give answers — ask probing questions: 'What do you notice about the arrangement of the vascular bundles in the sukuma wiki compared to the mango? What layer in the mango section do you think was responsible for making new wood?' [WAIT TIME: After each probe, wait at least 8 seconds before prompting further.] If groups are struggling to see the cambium, point to the thin ring between the bark and the wood in the mango section and say: 'Look carefully at this boundary zone. What might be special about this layer?' After the group activity, bring the class back and ask one spokesperson per | Structured observation with a comparison table forces students to look carefully at both specimens before receiving explanations. The reference diagram provides scaffolding without replacing discovery. Introducing vocabulary AFTER observation ensures words are attached to observed phenomena rather than abstract definitions. | Check observation tables for accuracy and detail — are students noting the ring-like arrangement of xylem in the woody stem? Are they noting that the sukuma wiki has scattered or loosely arranged vascular bundles? Listen for whether students can articulate that the mango section has a distinct boundary layer between bark and wood. Use these observations to gauge readiness for the Explain Phase. |

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| | notebooks with three columns: 'Feature', 'Sukuma wiki stem', 'Mango twig.' Features to compare include: colour, texture, presence or absence of rings, ease of bending, visible layers, and any other noticed features. After 5 minutes of observation, each group discusses: 'Which of these stems do you think can keep getting thicker year after year? What evidence from the specimen supports your answer?' | group to share one surprising observation. Record key observations on the board in a shared class table. Introduce key vocabulary ONLY after students have described what they see in their own words: say 'Scientists call this thin ring of tissue the vascular cambium — does that name help you remember its job, based on what you just observed?' | | |
| **Explain Phase**<br>📹 **VIDEO:**<br>Sinan, Rüstem Pasha Mosque<br>Source: Khan Academy (English - US curriculum)<br>🔍 Search ARES for similar videos<br><br>📖 **PDF:**<br>Orange-book-primary-2017<br>Source: Kenya Curriculum Tools 2025<br>🔍 Search ARES for similar readings | The teacher leads a whole-class explanation using a large diagram on the board (drawn or projected) showing: (1) a shoot tip with an apical meristem labelled, with arrows showing elongation downward; (2) a cross-section of a young stem; (3) a cross-section of an older woody stem with vascular cambium, secondary xylem (wood), secondary phloem, and bark labelled. Students follow along, adding labels to their own diagrams in their notebooks. The teacher explains primary growth using the relatable example of how sukuma wiki grows taller — the apical meristem at the tip divides and pushes the stem upward. The teacher then explains secondary growth using the mango tree: the vascular cambium, a ring of cells between the xylem and phloem, divides outward to produce new phloem and inward to produce new xylem (wood), making the trunk wider. Students are told that each growing season in a Kenyan mango tree produces one visible ring — so counting rings in a trunk cross-section can tell you the tree age. Students then answer three written questions individually: (1) What is the difference between | Begin the explanation by referring back to the predictions on the board: 'Let us now look at the evidence and see how it compares to what we predicted.' Use the phrase 'The evidence from our stems shows us...' to ground explanations in observation rather than abstract authority. When introducing vascular cambium, say: 'Remember that thin ring you noticed in the mango section? That is the vascular cambium. It is only one to a few cells thick, but it divides all year to produce new wood on the inside and new phloem on the outside — that is how the trunk grows wider.' Draw the cambium ring on the board and animate with arrows showing new cell production inward and outward. When explaining annual rings, say: 'In Kenya, mango trees grow faster in the long rains than in the dry season. The difference in cell size between fast-growth and slow-growth wood creates a visible ring. How many rings would a mango tree planted in 2010 have today?' [WAIT TIME: 10 seconds.] After students respond, ask: 'So what could you conclude about a mango tree with 15 rings?' [WAIT TIME: 8 seconds.] During the three | Concept clarification through diagram annotation connects abstract vocabulary to observed specimens. The three-question written reflection is a structured processing strategy — writing forces commitment to an explanation. Referencing the mango annual ring count question creates a concrete, locally meaningful application of the concept that anchors the explanation in Kenyan context. | The three written questions serve as the primary formative assessment for this phase. Check whether students: (1) correctly attribute primary growth to apical meristems and secondary growth to vascular cambium; (2) correctly distinguish increase in length from increase in girth; (3) correctly explain that herbaceous plants lack or have minimal vascular cambium activity. Collect two or three notebooks for spot-checking after the lesson. Use misconceptions identified to open Lesson 6. |

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| | primary growth and secondary growth? (2) Which type of meristem is responsible for each? (3) Why does a sukuma wiki stem not develop annual rings but a mango trunk does? After writing, pairs share answers before a class discussion. | written questions, circulate and read responses without correcting — identify one or two exemplary answers to share anonymously with the class during the discussion. | | |
| **Driving Question Board (DQB) Creation**<br>📹 **VIDEO:** [Sinan, Rüstem Pasha Mosque]<br>Source: Khan Academy (English - US curriculum)<br>🔍 Search ARES for similar videos<br><br>📖 **PDF:** [Orange-book-primary-2017]<br>Source: Kenya Curriculum Tools 2025<br>🔍 Search ARES for similar readings | The class returns to the Driving Question Board that has been built across Lessons 1 to 4. The teacher reads aloud any questions previously posted that relate to today's topic — for example, 'Why does the mango trunk get thicker every year while the bean plants stay thin?' and 'What is inside a tree trunk that is not inside a bean stem?' Students vote with a show of hands on whether each question has now been answered. For questions now answered, the class collaboratively writes a one-sentence evidence-based answer on a green sticky note and attaches it next to the original question. Students then individually write one new question on a yellow sticky note prompted by today's learning — for example, questions about what controls the rate of cambium division, whether all trees have annual rings, or whether growth hormones affect the cambium. New questions are added to the DQB. The teacher reads three of the new questions aloud and says: 'These are excellent — some of them are exactly what we will begin to explore in our next lesson on chemical messengers in plants.' | Before returning to the DQB, say: 'In Lesson 1 we were puzzled by the mango tree in the Kisumu farmer garden growing a thicker trunk year after year while the beans stayed thin. What would you say to that farmer now to explain what is happening inside the mango trunk?' [WAIT TIME: 10 seconds — allow students to think before calling on anyone.] Accept two or three responses and use them to frame the DQB update. When adding new questions, validate all contributions by saying: 'That is a question that scientists are still working on — it shows you are thinking like a researcher.' If time allows, ask: 'Is there anything from our predictions at the start of the lesson that turned out to be correct? What did the evidence change about your thinking?' This closes the prediction-evidence loop. | The DQB update ritual reinforces that science is a progressive process of building and revising knowledge. Colour-coding (green for answered, yellow for new questions) gives students a visual record of their growing understanding. Connecting new questions explicitly to the next lesson maintains narrative coherence across the unit storyline. | Read the new questions students post on the DQB to assess the quality of their scientific thinking. Are students asking mechanistic questions (how does the cambium know when to divide?) or only descriptive ones (what does cambium look like?)? Mechanistic questions indicate deeper conceptual engagement. Share exemplary new questions at the start of Lesson 6 to reward intellectual curiosity. |
| **Model Building Phase**<br>📹 **VIDEO:** | In the final 7 minutes of the lesson, students draw and annotate their own initial explanatory model in their notebooks under the heading: | Say: 'Scientists use models to show processes that are invisible to the naked eye. Your model should tell the story of what | Model construction externalises student thinking and reveals conceptual gaps that observation and explanation alone may not | Collect or photograph two to three student models at the end of class for review. Look for: correct placement of cambium between |

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| Sinan, Rüstem Pasha Mosque<br>Source: Khan Academy (English - US curriculum)<br>🔍 Search ARES for similar videos<br><br>📖 **PDF:** Orange-book-primary-2017<br>Source: Kenya Curriculum Tools 2025<br>🔍 Search ARES for similar readings | 'How a tree trunk grows thicker over time.' The model should include: (a) a cross-section circle representing a young stem with vascular cambium labelled; (b) arrows showing new xylem forming inward and new phloem forming outward from the cambium; (c) a second larger cross-section circle representing the same stem after several years of growth, with annual rings and thicker bark; (d) a brief caption (one to two sentences) explaining the process. Students who finish early add a comparison panel showing why the sukuma wiki cross-section does NOT show this pattern. The teacher circulates and uses the prompt: 'What would you add to your model if you wanted someone who had never studied plants to understand why the mango tree in the Kisumu farmer garden gets thicker every year?' | happens inside the trunk — not just show what it looks like.' Show a rough teacher-drawn example of before and after cross-sections on the board (incomplete — do not draw the cambium arrows yet, so students must add those details themselves). Circulate and give individual feedback using asset-based language: 'Your cambium ring is clearly labelled — now show me what moves outward from it and what moves inward.' [WAIT TIME: Give students at least 5 minutes of uninterrupted modelling time before circulating widely.] In the final 1 minute, ask two or three volunteers to hold up their models and describe one feature to the class. Close by saying: 'We will return to these models in Lesson 6 when we find out what chemical signals tell the cambium to start and stop dividing — and your models will need to be updated then.' | surface. The two-stage model (young stem and older stem) scaffolds the dynamic, process-based nature of secondary growth. The prompt about the sukuma wiki comparison reinforces the herbaceous versus woody distinction as a core explanatory contrast. | xylem and phloem; arrows showing directionality of new cell production; correct representation of xylem accumulating inward (forming wood) more than phloem outward; presence of annual ring lines in the older stem cross-section. Common gap to watch for: students drawing the whole stem expanding uniformly rather than new layers being added from the cambium ring. Use flagged models to open a targeted clarification at the start of Lesson 6. |

## D. TEACHER REFLECTION

After the lesson, consider the following questions: (1) Could students articulate the difference between primary and secondary growth in their own words, or were they reproducing vocabulary without understanding? (2) Did the physical specimens (sukuma wiki and mango cross-sections) create genuine cognitive engagement, or did students need more scaffolding to connect what they saw to the explanation? (3) Were the three written questions differentiated enough to assess a range of understanding? Consider adding a higher-order question for students who finish early: 'A botanist claims that all plants undergo primary growth but not all plants undergo secondary growth. Do you agree? Use evidence from today to justify your answer.' (4) Did the DQB update feel meaningful or procedural? If students struggled to generate new questions, consider providing sentence starters: 'I now wonder whether...' or 'Today made me curious about...' (5) How well did the lesson connect to the anchoring phenomenon? Students should leave with a clear explanation for why the mango and avocado trees in the Kisumu farmer garden have thick woody trunks while the bean plants remain thin — if this connection was not explicit, reinforce it at the start of Lesson 6.

## E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

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| **What did I observe?** | We observed cross-sections of a sukuma wiki stem and a mango twig. The sukuma wiki stem was green, soft, and showed loosely arranged vascular bundles with no visible rings. The mango twig was brown and hard, showed a distinct ring between the bark and the inner wood, and had visible growth rings in the woody core. |

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| **What did I learn?** | We learned that plants grow in two distinct ways: primary growth increases the length of roots and shoots through cell division in apical meristems at the tips of roots and shoots; secondary growth increases the girth (thickness) of stems and roots through cell division in the vascular cambium, a ring of lateral meristem tissue located between the xylem and phloem. Each growing season, the vascular cambium of a Kenyan mango or avocado tree produces a new layer of xylem inward and phloem outward, creating an annual ring. Herbaceous plants such as sukuma wiki, beans, and maize have little or no vascular cambium activity and therefore do not develop a thick woody trunk. |
| **How does this explain the phenomenon?** | We can now explain why the mango and avocado trees next to the Kisumu farmer bean plants grow thicker trunks every year while the bean plants remain thin: the trees have an active vascular cambium that produces new wood (secondary xylem) each season, while the bean plants rely almost entirely on apical meristem activity for growth in length and have no significant secondary growth. This also explains why counting the annual rings in a mango trunk cross-section can reveal the age of the tree, and why timber from Kenyan trees is dense and strong — it is the accumulated product of many years of secondary growth from the vascular cambium. |

| LESSON 6 (40 minutes): Chemical Messengers in Plants: Growth Hormones | |
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| **A. SPECIFIC LEARNING OUTCOMES** | |
| **Category** | **Specific Learning Outcomes** |
| **Purpose** | Students will investigate the role of plant growth hormones in regulating growth, development, dormancy, and responses to the environment, connecting chemical signals to observable plant behaviours seen in the anchoring phenomenon. |
| **Knowledge** | Students will be able to: (1) identify the five major plant hormones — auxins, gibberellins, cytokinins, abscisic acid, and ethylene — and their primary functions; (2) explain how auxin redistribution causes phototropism and apical dominance; (3) connect gibberellins to stem elongation and seed germination; (4) explain how abscisic acid promotes seed dormancy; (5) link ethylene to fruit ripening in Kenyan crops such as mangoes and bananas. |
| **Skills** | Students will practise: using print and digital media to research scientific information; constructing and interpreting labelled diagrams of hormone action; designing a simple phototropism demonstration; updating an explanatory model with new evidence about chemical regulation. |
| **Attitudes** | Students will appreciate that invisible chemical signals — not just physical conditions — control how plants grow and respond to their environment, developing curiosity about biochemical processes and their agricultural applications in Kenya. |
| **Key Inquiry Question** | Why is the concept of growth in plants important? |
| **Purpose in Storyline** | Lesson 6 adds the chemical regulation layer to the class explanatory model. Students can now explain not only HOW plants grow in length and girth (Lesson 5) but WHY growth is directed — toward light, away from competitors, held in pause during drought — through hormone signals. This directly explains the spindly pale sukuma wiki grown in low light, the dormant seeds from Lesson 2, and the differential emergence patterns from the anchoring phenomenon. The DQB gains a new cause-and-effect column connecting hormones to observed plant behaviours. |
| **Safety Notes** | No chemical hazards in this lesson. If using potted seedlings for the phototropism demonstration, handle pots carefully to avoid soil spills. Students with plant allergies should be seated away from any live plant material. Wash hands after handling soil or plant material. |

| B. LESSON OVERVIEW |
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| In this 40-minute lesson students move from understanding the structural mechanisms of growth (Lesson 5) to understanding the chemical control of growth. Using print resources, a class-constructed hormone function chart, and a short video clip or diagram analysis, students discover how five plant hormones — auxins, gibberellins, cytokinins, abscisic acid, and ethylene — orchestrate plant growth and development. The supporting phenomenon of spindly sukuma wiki growing toward a window is used to anchor the concept of phototropism and auxin redistribution. Students connect abscisic acid back to dormancy (Lesson 2) and ethylene to the ripening of mangoes and bananas familiar in Kenyan markets. The lesson closes with students updating the class DQB and adding a hormone layer to their individual explanatory models, preparing them for the environmental factors lesson that follows. |

| C. LESSON IMPLEMENTATION FRAMEWORK | | | | |
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| **Phase** | **Learner Experience** | **Teacher Moves** | **Sensemaking Strategy** | **Formative Assessment Strategy** |
| **Predict Phase** | Students open their notebooks to a | Display the three scenarios as | Activate prior knowledge by linking | Collect quick-write notebooks at |

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| 📹 **VIDEO:** [Balancing more complex chemical equations]<br>Source: Khan Academy (English - US curriculum)<br>🔍 Search ARES for similar videos<br><br>📖 **HTML:** [Auxins - Advanced (Flexbook)]<br>Source: CK-12<br>🔍 Search ARES for similar readings | quick-write prompt displayed on the board: 'A pot of sukuma wiki growing near a window bends toward the light over five days. A seed in dry soil does not germinate even though it has water nearby. A mango in Kisumu market ripens faster when placed next to a banana. What do these three observations have in common?' Students write individually for 2 minutes without talking, then share with a shoulder partner. The teacher collects 4 to 5 verbal responses and writes key student words on the board — expected answers include: the plant can sense things, something inside controls it, chemicals, signals. Teacher validates all responses and frames the lesson: 'Today we are going to discover that plants make their own chemical messengers called hormones that control all of these things — and more.' | images if possible: a bent sukuma wiki stem, a dry dormant seed, mangoes and bananas together on a market table in Kisumu. Ask: 'What single idea could explain all three of these?' WAIT 30 SECONDS before taking responses. Use the move: 'Do not tell me the answer yet — just tell me what the three situations share.' After collecting responses, say: 'You are already thinking like plant biologists. Scientists found the answer in the 1920s and it changed agriculture forever. Let us find out what they discovered.' Write the word HORMONES on the board next to student suggestions that pointed in that direction, validating student thinking. | to human hormones if students have studied biology: 'You may know that humans have hormones like adrenaline that tell the body to react quickly. Plants have their own version.' This analogical bridge lowers the cognitive load of the new concept and primes students to think of hormones as internal messengers rather than external conditions. | the start and scan for misconceptions — particularly the idea that plants consciously choose to bend or that ripening is caused only by temperature. Note which students connect any of the three scenarios to an internal chemical cause, as these students can be called on during the Explain phase to scaffold peers. |
| **Observe Phase**<br>📹 **VIDEO:** [Balancing more complex chemical equations]<br>Source: Khan Academy (English - US curriculum)<br>🔍 Search ARES for similar videos<br><br>📖 **HTML:** [Auxins - Advanced (Flexbook)]<br>Source: CK-12<br>🔍 Search ARES for similar readings | Students work in groups of four. Each group receives a one-page printed resource card (teacher-prepared or from the textbook) listing the five plant hormones with a short description and one Kenyan example for each. Groups also receive a blank five-column table to complete. If a device is available, one group may use a short curated video clip (2 to 3 minutes) showing phototropism in a seedling. Groups read, discuss, and fill in the table: Hormone name — Where it is mainly produced — Main function — What happens without it or in excess — Kenyan crop or plant example. Teacher circulates, pausing at each group for 30 seconds to listen to discussion without intervening | As students work, pose probing questions group by group: 'Your table says auxin causes bending — but WHERE in the plant is the auxin moving and WHY does that cause bending?' WAIT 20 SECONDS. If no response: 'Think about what auxin does to individual cells — does it make them grow more or less?' Use the potted seedling as a focal object: 'Look at this sukuma wiki. Where is the light coming from? Which side of the stem is longer? What does that tell you about where the auxin went?' Do not give the answer — let students generate the explanation. Say: 'Keep that observation in your mind. We will use it in the next phase.' | The five-column resource table serves as a structured observation scaffold, preventing students from simply copying text without thinking. The Kenyan crop column forces students to localise each hormone — ethylene in ripening mangoes and bananas in Mombasa markets, gibberellins in the elongation of sugarcane in Mumias, abscisic acid in dormant sorghum seeds stored by Turkana farmers. This contextualisation builds meaningful schema rather than isolated vocabulary. | Circulate and check that the Kenyan crop column is being completed with specific, reasoned examples rather than left blank. Ask any group that writes only hormone names: 'Can you think of a plant you have seen in your village or market that shows this behaviour?' This reveals whether students are connecting textbook concepts to real experience. Collect one table per group at the end of the phase to check for accuracy before the Explain phase. |

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| | unless a group is stuck. A potted seedling that has been placed near a one-sided light source for two to three days (set up before the lesson) is displayed at the front of the room. Students are asked to observe it and sketch what they see before the explanation is given. | | | |
| **Explain Phase**<br>📹 **VIDEO:** [Balancing more complex chemical equations]<br>Source: Khan Academy (English - US curriculum)<br>🔍 Search ARES for similar videos<br><br>📖 **HTML:** [Auxins - Advanced (Flexbook)]<br>Source: CK-12<br>🔍 Search ARES for similar readings | The class reconvenes. The teacher draws a large stem diagram on the board showing a seedling with a light source on one side. Students are called on to explain step by step what happens to auxin: (1) auxin is produced at the shoot tip; (2) light causes auxin to migrate to the shaded side; (3) cells on the shaded side elongate more; (4) the stem curves toward the light. Students label a diagram in their notebooks. The teacher then introduces apical dominance: 'Why does the main shoot of a maize plant grow taller while side shoots stay short? Auxin from the top suppresses lateral buds.' Students connect this to the branching patterns they observed in beans versus maize during the anchoring phenomenon. Next, students map abscisic acid back to Lesson 2: 'We said some seeds stay dormant because of chemical inhibitors — that inhibitor is abscisic acid. When water washes it away or temperature changes, dormancy breaks.' Finally, students write one sentence each linking gibberellins to stem elongation, cytokinins to cell division, and ethylene to the mango and banana observation from the Predict phase. | Use the Socratic questioning sequence: 'If auxin makes cells elongate and it moves to the shaded side — what will happen to the cells on the shaded side versus the lit side?' WAIT 30 SECONDS. 'If the shaded side cells are longer — which way will the stem bend?' WAIT 20 SECONDS. If students are uncertain, say: 'Imagine a balloon inflated more on one side — which way does it curve?' After explaining apical dominance, ask: 'If you were a Kenyan tea farmer in Kericho and you wanted your tea bushes to branch out wide and be bushy — what would you do to the main shoot based on what you now know about auxin?' WAIT 30 SECONDS. Expected answer: prune the tip — removing the source of auxin so lateral buds can grow. Affirm and connect to real agricultural practice: 'That is exactly what tea pruning does — it is applied plant hormone science.' | The diagram-building sequence (stem, light arrow, auxin migration arrow, cell elongation comparison, bend direction) transforms an abstract concept into a spatial, visual causal chain. This is a 'mechanism mapping' strategy — students are not memorising a fact but tracing a mechanism step by step, which supports transfer to new situations such as the driving question performance task in Lesson 8. | Ask students to hold up their notebooks showing their labelled auxin-phototropism diagram. Scan for: correct direction of auxin movement, correct side of stem elongation, correct direction of bending. Common error to watch for: students drawing auxin moving toward the light instead of away from it. Address immediately with the question: 'Which side of this plant stem is longer — the shaded or the lit side? So where must the auxin be?' |
| **Driving Question** | Students return to the class DQB displayed at the front of the room. | Read the unanswered DQB questions slowly: 'Here is a | The DQB annotation ritual — physically moving question cards | Collect the new sticky-note questions and sort them into three |

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| **Board (DQB) Creation**<br>📹 **VIDEO:**<br>Balancing more complex chemical equations<br>Source: Khan Academy (English - US curriculum)<br>🔍 Search ARES for similar videos<br><br>📖 **HTML:**<br>Auxins - Advanced (Flexbook)<br>Source: CK-12<br>🔍 Search ARES for similar readings | The teacher reads aloud questions that were posted in earlier lessons and asks: 'Which of these questions can we now answer using what we know about plant hormones?' Students nominate questions and suggest answers. Expected connections: 'Why do some seeds not germinate even with water?' — abscisic acid maintains dormancy (Lesson 2 question now answered by hormone knowledge). 'Why does sukuma wiki in the window grow toward the light?' — auxin redistribution. 'Why do bean plants branch more than maize?' — apical dominance and auxin. Students write two NEW questions generated by today's lesson on sticky notes and place them in the 'What we still wonder' column. Expected new questions: 'Can farmers use hormones to make plants grow faster?' 'Does every plant have the same hormones?' 'What happens to a plant if it makes too much auxin?' | question someone wrote in Lesson 2: Why do chemical inhibitors stop a seed from germinating? Can anyone connect that question to something we learned today?' WAIT 30 SECONDS. After a student responds, ask the class: 'Does everyone agree? What is the hormone involved?' Then say: 'Let us move that question card to the answered column and write the answer underneath it.' Model annotating answered questions with evidence. For new questions, say: 'What NEW mysteries did today open up for you? Write one question you genuinely want to know — there are no wrong questions here.' | from unanswered to answered and writing evidence-based answers beneath them — makes student learning progress visible and tangible. It also demonstrates that science is a cumulative enterprise: each lesson answers some questions and generates new ones, mirroring the nature of scientific inquiry. | categories: questions answerable in Lessons 7 or 8 (agricultural applications, environmental factors), questions that extend beyond the curriculum (genetic engineering, synthetic hormones), and misconception questions (for example, 'Do plants feel pain from hormones?'). Use the misconception questions as warm-up prompts in the next lesson. |
| **Model Building Phase**<br>📹 **VIDEO:**<br>Balancing more complex chemical equations<br>Source: Khan Academy (English - US curriculum)<br>🔍 Search ARES for similar videos<br><br>📖 **HTML:**<br>Auxins - Advanced (Flexbook)<br>Source: CK-12<br>🔍 Search ARES for similar readings | Students open their individual explanatory model pages (started in Lesson 5) and add a new layer: a hormone regulation overlay. They draw arrows and labels showing: (1) auxin at the shoot apex controlling phototropism and apical dominance; (2) abscisic acid in the seed coat linked to the dormancy box they drew in earlier lessons; (3) gibberellins in the elongating stem; (4) ethylene in ripening fruit. They connect the hormone diagram to their existing primary and secondary growth diagrams using coloured pencils if available — one colour for structural growth mechanisms, another for chemical control | Say: 'Your model now has two systems working together — a structural system (meristems, cell division, xylem, phloem) and a chemical control system (hormones). Use two different colours to show these are different but connected systems.' Circulate and prompt students who are drawing hormones floating randomly: 'Where exactly is auxin produced? Draw it there. Where does it move to in phototropism? Draw that arrow.' For the class model, call on two or three students to come forward and add their hormone arrows to the shared diagram, saying: 'Show us where you put abscisic acid and | The two-colour model overlay strategy makes the distinction between structural and chemical regulation explicit and spatial. Students who struggle with the abstract idea of hormones are anchored by the physical locations on their stem diagrams — hormone action becomes associated with specific places (shoot tip, seed coat, ripening fruit) rather than floating abstractions. This prepares students for the environmental factors lesson by giving them a model that can absorb new inputs (light, water, temperature affecting hormone production). | Before students close notebooks, ask them to write one sentence completing: 'Before today I thought plants grew only because of... but now I understand that...' Collect these exit-ticket sentences. Review them before Lesson 7 to identify students who still hold purely structural or purely environmental explanations without integrating hormones. These students will need targeted support during the Lesson 7 group activity where all factors — structural, hormonal, and environmental — are synthesised. |

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| | signals. Students then write one sentence at the bottom of their model page answering: 'How does adding hormones change your understanding of why the Kisumu farmer saw such different plant behaviours?' The class model on the board is updated by the teacher with student contributions, adding a hormone regulation layer in a new colour. | explain to the class why you put it there.' WAIT 20 SECONDS after each student addition before asking: 'Does anyone want to add to or adjust what was just drawn?' | | |

## D. TEACHER REFLECTION

After the lesson, consider: Did students successfully connect abscisic acid back to dormancy from Lesson 2, or did they treat it as an isolated new fact? Could students explain the mechanism of phototropism step by step, or did they memorise the outcome without understanding the auxin migration process? Were the Kenyan crop examples (mango ripening, Kericho tea pruning, Mumias sugarcane elongation) genuinely used by students to anchor the concepts, or were they superficial add-ons? Which DQB questions generated the most discussion — and what does that reveal about student curiosity and gaps? For Lesson 7, identify two or three students whose exit tickets showed strong integrative thinking and seat them as group leaders for the regional crop case studies.

## E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

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| **What did I observe?** | A sukuma wiki stem bending toward a window; a potted seedling tilted toward a one-sided light source; mangoes and bananas ripening quickly when stored together in a Kisumu market; dormant seeds that refuse to germinate even when given water and warmth. |
| **What did I learn?** | Plants produce five major hormones: auxins control phototropic bending and apical dominance by causing unequal cell elongation; gibberellins promote stem elongation and help break seed dormancy; cytokinins stimulate cell division and lateral bud growth; abscisic acid maintains seed dormancy and closes stomata during drought; ethylene triggers fruit ripening and leaf drop. These chemical messengers work by signalling specific cells to grow, divide, or stop, without the plant having a nervous system. |
| **How does this explain the phenomenon?** | The Kisumu farmer saw the bean seedlings push upward and arch while maize shot straight — differences shaped by germination type and apical dominance controlled partly by auxin. The dormant seeds that refused to germinate despite water and warmth were held by abscisic acid acting as a chemical lock. The mangoes in the market ripened faster near bananas because ethylene released by one fruit triggered ripening in the other. The spindly sukuma wiki near a dim window elongated toward light because auxin migrated to the shaded side, causing those cells to grow longer and bend the stem toward the light source. |

## LESSON 7 (40 minutes): Factors Favouring Plant Growth: Environmental Influences and Cumulative Model Building

### A. SPECIFIC LEARNING OUTCOMES

| Category | Specific Learning Outcomes |
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| **Purpose** | Students investigate how environmental factors — water, temperature, light, and nutrients — interact with the biological mechanisms of germination, growth types, and hormones to determine how and where plants thrive in Kenya. |
| **Knowledge** | Students will be able to: (1) identify and explain the role of water, temperature, light, and nutrients as factors that influence plant growth and development; (2) describe how specific Kenyan crop systems — tea in Kericho, maize in the Rift Valley, and mangroves in Mombasa — reflect adaptations to distinct environmental conditions; (3) connect environmental factors to previously learned concepts including germination conditions, primary and secondary growth, and plant hormones. |
| **Skills** | Students will analyse regional case studies using structured reading and group discussion; interpret data on crop growth conditions; present findings to the class using evidence-based reasoning; and contribute to a cumulative class explanatory model that integrates all sub-strand concepts. |
| **Attitudes** | Students will appreciate the complexity of plant growth as shaped by interacting biological and environmental factors, and develop curiosity about how Kenyan farmers and agronomists manage these factors to improve food security. |
| **Key Inquiry Question** | Why is the concept of growth in plants important? — Students now extend this question to environments: growth is important because it is shaped by and adapted to specific environmental conditions, meaning farmers must understand these factors to support healthy crops. |
| **Purpose in Storyline** | This is Lesson 7 of 8 and serves as the integrative evidence-gathering lesson. Students now bring together all prior knowledge — dormancy, germination conditions, epigeal and hypogeal germination, primary and secondary growth, and hormones — and connect these mechanisms to the real-world environmental contexts that shape plant outcomes. This lesson directly advances the Driving Question by adding the final explanatory layer: the Kisumu farmer sees different outcomes not just because of seed biology, but because environmental factors interact with that biology to produce the plants we observe. |
| **Safety Notes** | No hazardous materials are used in this lesson. If printed case study cards are used, ensure legibility for all learners. If the lesson is conducted outdoors or near a school garden for observation, remind students not to uproot or damage plants. Students with allergies to dust or pollen should be noted if plant specimens are brought indoors. |

### B. LESSON OVERVIEW

In a 40-minute lesson, students work in three expert groups, each analysing a Kenyan regional crop case study: Group A studies tea farming in Kericho (cool temperatures, high rainfall, shade), Group B studies maize farming in the Rift Valley (warm temperatures, seasonal rainfall, high light), and Group C studies mangrove plants in Mombasa (saline water, high humidity, tidal zones). Each group identifies which environmental factors — water, temperature, light, and nutrients — are critical in their region and how plants are adapted or managed accordingly. Groups share findings with the class. The class then collaborates to build a cumulative explanatory model on the board that connects germination, growth types, hormones, and environmental factors back to the anchoring phenomenon of the Kisumu farmer. The DQB is updated to reflect newly answered and newly arising questions.

### C. LESSON IMPLEMENTATION FRAMEWORK

| Phase | Learner Experience | Teacher Moves | Sensemaking Strategy | Formative Assessment Strategy |
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| **Predict Phase**<br>📹 **VIDEO:**<br>Energy inputs for tilling a hectare of land<br>Source: Khan Academy (English - US curriculum)<br>🔍 Search ARES for similar videos<br><br>📖 **HTML:**<br>Adding Mixed Fractions of Water (PLIX)<br>Source: CK-12<br>🔍 Search ARES for similar readings | Students begin seated. The teacher displays or reads aloud three short descriptions on the board: (1) Tea plants in Kericho grow slowly, have dense dark-green leaves, and thrive at altitudes above 1500 m in cool, misty conditions. (2) Maize in the Rift Valley grows quickly during the long rains, is planted in open fields with full sun, and yellows when nitrogen is scarce. (3) Mangrove trees in Mombasa grow in salty tidal water, have thick waxy leaves, and develop prop roots that stick out of the mud. The teacher asks: 'Before we investigate these plants properly, write down in your notebook: which ONE environmental factor do you think is MOST important for each crop, and WHY?' Students write silently for 2 minutes. The teacher then asks three volunteers — one per scenario — to share their prediction. The teacher does NOT evaluate predictions at this stage but records them visibly on a corner of the board under the heading 'Our Predictions.' | Teacher reads each description clearly and points to a map of Kenya showing Kericho, the Rift Valley, and Mombasa so students have geographic context. Teacher says: 'You have seen these regions on maps and perhaps heard of them. Use what you already know about Kenya and what we have been learning about plant growth to make your best prediction. There is no wrong answer right now — we are thinking scientists.' Teacher gives 2 minutes of silent writing time and enforces quiet so all students engage independently. After sharing, teacher says: 'Hold those predictions. Let us see what the evidence says.' WAIT TIME: After asking each student to share, teacher waits 5 seconds before prompting or moving on, allowing the student to complete their thought. | Activating prior knowledge through prediction connects new environmental content to students existing understanding of germination conditions from Lesson 3 and hormones from Lesson 6. Anchoring predictions on the board creates accountability — students will return to them during the Explain Phase to assess accuracy. | Teacher listens to verbal predictions and scans written notes to check whether students are drawing on prior sub-strand concepts (for example, referencing water as a germination requirement from Lesson 3, or temperature affecting hormone activity from Lesson 6). If students only give surface-level answers such as 'water is important for all plants,' teacher probes: 'Can you connect that to something specific we learned about how water affects a plant cell or a seed?' |
| **Observe Phase**<br>📹 **VIDEO:**<br>Energy inputs for tilling a hectare of land<br>Source: Khan Academy (English - US curriculum)<br>🔍 Search ARES for similar videos<br><br>📖 **HTML:**<br>Adding Mixed Fractions of Water (PLIX) | Students form three groups. Each group receives a Case Study Card (one side contains descriptive text and data; the other side contains 3 to 4 guiding questions). Group A — Tea in Kericho: The card describes how tea is grown at 1500 to 2700 m altitude, mean temperatures of 14 to 26 degrees Celsius, annual rainfall of 1200 to 1400 mm, shade trees used to diffuse light, and how tea bushes are pruned to maintain lateral growth. Guiding questions include: Which factor do you think limits tea | Teacher circulates to each group during the 10-minute group work period. For Group A, if students are stuck on the pruning-apical dominance connection, teacher says: 'Think back to Lesson 6 — when the apical bud is removed, which hormone balance changes and what starts to grow?' For Group B, if students mention nitrogen without explaining the mechanism, teacher says: 'Nitrogen is in amino acids and chlorophyll — what does that mean for the rate of cell division | The case study card structure (text plus guiding questions plus evidence table) scaffolds structured inquiry. The guiding questions are deliberately written to force connections to prior lessons, ensuring this is not isolated factual recall but integrative sensemaking. The three Kenyan contexts are chosen to represent diversity across the country — highland, savanna, and coastal — making the lesson culturally and geographically relevant. | Teacher checks each group's evidence table for accuracy and depth of connections. A strong response for Group B might read: 'Nitrogen — needed for chlorophyll production — without it, leaves turn yellow and photosynthesis slows, reducing sugar supply for cell division and growth. Connected to Lesson 5: less sugar means less material for secondary growth.' A weak response would read: 'Nitrogen is important for maize.' Teacher uses this to calibrate the level of support |

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| Source: CK-12<br>🔍 Search ARES for similar readings | growth at lower, hotter altitudes? How does pruning relate to what we learned about apical dominance? Group B — Maize in the Rift Valley: The card describes how maize requires 18 to 30 degrees Celsius, full sun for photosynthesis, well-drained soils with nitrogen and phosphorus, and how yellow leaves in maize often signal nitrogen deficiency. Guiding questions include: What happens to photosynthesis and growth if maize is planted too close together and shaded? How does water availability link to the germination conditions we tested in Lesson 3? Group C — Mangroves in Mombasa: The card describes how mangroves tolerate salt through special root structures, have thick cuticles to reduce water loss despite being surrounded by water, produce prop roots to access oxygen in anaerobic mud, and grow slowly with dense secondary growth. Guiding questions include: Why might nutrient levels in tidal mud affect mangrove height? How does secondary growth (from Lesson 5) help mangroves survive physical stress from tides? Students read, discuss within their group, and record key findings in a table with columns: Environmental Factor — How It Affects Growth — Connection to Previous Lessons. | and leaf colour?' For Group C, if students are puzzled by salt and water loss, teacher says: 'Can a plant take up water by osmosis if the outside water is saltier than the cell sap? What does that do to the plant?' WAIT TIME: After each probing question, teacher steps back and waits 8 seconds, allowing the group to discuss before answering. Teacher does not answer for them. | | during the Explain Phase presentations. |
| **Explain Phase**<br>📹 **VIDEO:** Energy inputs for tilling a hectare of land<br>Source: Khan Academy (English - US curriculum)<br>🔍 Search ARES for similar videos | Each group appoints a spokesperson who presents their case study findings to the class in 2 minutes. During each presentation, students from the other two groups listen and record one new thing they learned and one question they still have. After all three presentations, the teacher | Teacher manages time: 2 minutes per group presentation plus 30 seconds for one clarifying question from the audience. Teacher models a clarifying question for the first group to demonstrate the norm: 'You said tea grows better in cool temperatures — can you say more about what happens to the | Presentation followed by peer note-taking develops scientific communication skills and active listening. The three synthesis questions are specifically designed to force integration: question 1 surfaces water as universal; question 2 generates debate and reasoning; question 3 explicitly | Teacher listens for whether students can articulate a mechanistic connection between an environmental factor and a plant process — not just correlation. For example, a surface-level response would be: 'Light is important for all three.' A mechanistic response would be: |

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| 📖 **HTML:** [Adding Mixed Fractions of Water (PLIX)] Source: CK-12 🔍 Search ARES for similar readings | facilitates a whole-class discussion using three synthesis questions written on the board: (1) Which environmental factor appeared in ALL THREE case studies, even in different ways? (2) Which factor surprised you most — why does it matter so much? (3) How do any of these environmental factors work THROUGH the hormones we studied in Lesson 6? Students discuss in pairs for 1 minute before sharing whole-class. The teacher records key points in a summary column on the board next to the earlier predictions, and students compare: Were our predictions correct? What did we miss? | enzyme reactions involved in photosynthesis when it gets too hot?' After all presentations, teacher says: 'Now look at our predictions from the beginning of the lesson. With your partner, decide: which prediction was most accurate, and which one needs the biggest revision? You have 60 seconds.' After sharing, teacher says: 'This is exactly what scientists do — they predict, investigate, and then revise. Your revised thinking is MORE scientific than your first guess, not less.' WAIT TIME: After posting the three synthesis questions, teacher waits 10 seconds in silence before asking pairs to discuss, giving students time to read and process all three questions before talking. | links to Lesson 6 hormones, building the explanatory chain. Comparing predictions to findings closes the predict-observe-explain loop and builds metacognitive awareness. | 'Without enough light, plants cannot produce enough sugar through photosynthesis, so cell division and elongation slow down, and auxin redistribution during phototropism would cause bending toward any available light source.' Teacher notes which students are making mechanistic connections and which are still at descriptive level, to target support in Lesson 8 synthesis. |
| **Driving Question Board (DQB) Creation** 📹 **VIDEO:** [Energy inputs for tilling a hectare of land] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos<br><br>📖 **HTML:** [Adding Mixed Fractions of Water (PLIX)] Source: CK-12 🔍 Search ARES for similar readings | The teacher directs students to return to the class Driving Question Board displayed at the front. The teacher reads aloud the original Driving Question: 'Why do seeds from the same garden grow and develop so differently, and what controls how a plant grows from a tiny seed into a tall tree or a flowering herb?' Students are given sticky notes (or asked to write in their notebooks if sticky notes are unavailable). Each student writes one answer or partial answer they can NOW give to the Driving Question — something they could not have said before today or before this unit. Students post or read their responses. The teacher then asks: 'Are there any NEW questions this lesson has raised that we have not yet answered?' Students share. Teacher marks already-answered questions on the DQB with a tick and adds new questions. The | Teacher points to specific sticky notes or written entries from Lessons 1 and 2 on the DQB and reads them aloud: 'In Lesson 1 someone wrote: Why do some plants get big and woody while others stay small? Can someone now give an answer to that using what we know from Lessons 5, 6, and 7?' Teacher pauses and selects a student. After the student answers, teacher asks: 'Does anyone want to add to or challenge that explanation?' This models scientific discourse. Teacher then says: 'By next lesson, every question on this board should have at least a beginning of an answer. Look at what is still unanswered and think about what we still need.' WAIT TIME: After asking students to write their DQB response on their sticky note, teacher waits 90 seconds in complete silence to allow all students to write a | The DQB serves as a visible, cumulative record of the class inquiry journey. Returning to it in every lesson builds coherence across the eight-lesson sequence and helps students experience science as a progressive explanation-building process. Ticking off answered questions gives students a concrete sense of intellectual progress, which supports motivation and metacognition. | Teacher reads student DQB entries to assess the sophistication of explanations. A high-quality entry might read: 'Environmental factors like water and light work through hormones like auxins and gibberellins to control how fast and in which direction a plant grows — this is why the Kisumu farmer sees different results even from the same garden.' A low-quality entry would read: 'Plants need water and sunlight.' Teacher uses these entries to identify students who need additional scaffolding during the model-building phase and in Lesson 8. |

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| | teacher highlights one unanswered or partially answered question to signal what Lesson 8 will resolve: 'We still need to put ALL of this together into one complete explanation. That is what our final lesson will do.' | substantive response before posting. | | |
| **Model Building Phase**<br>📹 **VIDEO:** [Energy inputs for tilling a hectare of land]<br>Source: Khan Academy (English - US curriculum)<br>🔍 Search ARES for similar videos<br><br>📖 **HTML:** [Adding Mixed Fractions of Water (PLIX)]<br>Source: CK-12<br>🔍 Search ARES for similar readings | In the final 8 minutes, the class works together to build a cumulative explanatory model on the board. The teacher draws a large blank diagram frame with the title: 'How a Plant Grows: From Seed to Tree or Herb.' The frame has six labelled zones: (1) Dormant Seed, (2) Conditions for Germination, (3) Type of Germination, (4) Primary and Secondary Growth, (5) Hormones, (6) Environmental Factors. The teacher asks volunteers to come to the board and draw arrows or write labels connecting any two zones, explaining the connection aloud as they do. For example, a student might draw an arrow from Environmental Factors to Conditions for Germination and write: 'Water and warmth break dormancy and trigger germination.' Another student might connect Hormones to Primary Growth and write: 'Gibberellins and auxins cause cell elongation in apical meristems.' The teacher facilitates until at least 6 to 8 connections are drawn. Students copy the class model into their notebooks, adding any additional connections they can think of. The teacher closes by explicitly connecting the model back to the anchoring phenomenon: 'So — why did the Kisumu farmer see maize and beans emerge so differently, and why do his mango trees grow wide and tall while his beans stay thin? | Teacher scaffolds the model building by starting with the first connection themselves as a demonstration: 'I will draw the first arrow: from Dormant Seed to Conditions for Germination, because we learned in Lessons 2 and 3 that a seed stays dormant until water, oxygen, and warmth are present. Who can draw the NEXT connection?' Teacher selects students who have not yet contributed in the lesson to ensure broad participation. If a student draws an incorrect or unclear connection, teacher says: 'Interesting — class, do we agree with this arrow? Does anyone want to revise or add to it?' This distributes evaluation to the class rather than positioning the teacher as sole authority. Teacher ends by saying: 'This model is not finished. Lesson 8 is where you will complete it, individually, and use it to solve a real farmer problem from Nakuru. Think tonight: what connections are still missing?' WAIT TIME: After asking 'Who can draw the next connection?' teacher waits 7 seconds before selecting a student, giving thinking time and signalling that the question is not trivial. | Public model building makes student thinking visible and creates a shared artefact that encodes the class collective understanding. The six-zone framework ensures all major sub-strand concepts are represented and connected, preventing isolated recall. Copying the model into notebooks creates a personal record that students will revise in Lesson 8. Explicitly returning to the anchoring phenomenon at the close of the model-building phase reinforces that the entire inquiry has been in service of explaining one puzzling real-world observation. | Teacher assesses the quality of connections drawn during model building by asking contributing students to explain the mechanism behind each arrow, not just the label. A strong contribution includes cause-and-effect language: 'Environmental temperature affects the rate of enzyme activity in the seed, which controls whether germination can begin.' A weak contribution simply restates a label: 'Temperature goes to germination.' Teacher notes which conceptual connections are missing from the class model — for example, if no student connects ABA to dormancy, or if secondary growth is left isolated — and flags these as priorities for the Lesson 8 synthesis task. |

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| | Your model now contains the full answer. Tomorrow, you will complete it and use it to explain a new farmer problem.' | | | |

| D. TEACHER REFLECTION |
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| After the lesson, consider: Did all three groups successfully make at least one connection between their environmental case study and a mechanism from a previous lesson (dormancy, germination, growth type, or a hormone)? Were students able to articulate causal connections on the DQB or did most responses remain descriptive? Did the cumulative model on the board contain connections across all six zones, or were some zones left isolated? If the hormone zone or the secondary growth zone received few connections, plan to open Lesson 8 with a targeted prompt that asks students to revisit those zones first. Consider whether the three case studies — Kericho, Rift Valley, Mombasa — generated genuine discussion or whether one group struggled significantly. If Group C (mangroves) was the most challenging, consider providing a brief teacher-read summary of mangrove adaptation at the start of Lesson 8 before students begin independent model revision. Reflect also on participation equity: were students who have been quieter in previous lessons drawn into the model-building phase? Use the DQB entry quality as a pre-assessment for the Lesson 8 performance task scaffolding needed. |

| E. SUMMARY TABLE PROMPT (pre-filled example for this lesson) | |
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| **What did I observe?** | We observed how tea plants in cool Kericho, maize in the warm Rift Valley, and mangroves in saline Mombasa each grow under very different environmental conditions — and that water, temperature, light, and nutrients shape not just whether a plant grows, but how it grows, how fast, and in what form. |
| **What did I learn?** | We learned that environmental factors do not act independently — they interact with the biological systems inside the plant, including the conditions needed for germination, the type of growth (primary or secondary), and the action of hormones such as auxins, gibberellins, and abscisic acid, to produce the plant forms we see in different regions of Kenya. |
| **How does this explain the phenomenon?** | We can now explain more completely why the Kisumu farmer sees such different outcomes: the maize and bean seeds differ in germination type and hormone responses, the mango tree undergoes secondary growth driven by cambium activity, and all of these biological processes are shaped and modified by the water, temperature, light, and nutrients available in the garden environment. Plants are not growing in isolation — they are growing in conversation with their environment. |

| LESSON 8 (40 minutes): Putting It All Together: Synthesis, Cumulative Model Revision, and Assessment | |
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| **A. SPECIFIC LEARNING OUTCOMES** | |
| **Category** | **Specific Learning Outcomes** |
| **Purpose** | Students synthesise all learning from Lessons 1 through 7 to construct a complete explanatory model of plant growth and development, revisit the anchoring phenomenon and Driving Question Board, and demonstrate their understanding through a structured performance task. |
| **Knowledge** | Students can trace a seed journey from dormancy through germination (epigeal and hypogeal), primary growth, and secondary growth, explaining the role of growth hormones and environmental factors at each stage. |
| **Skills** | Students revise and annotate an explanatory diagram, apply integrated knowledge to a novel agricultural scenario, and communicate scientific reasoning in writing or orally. |
| **Attitudes** | Students demonstrate intellectual honesty by acknowledging what they did not initially understand, show confidence in using scientific evidence to advise a farmer, and appreciate the relevance of plant biology to Kenyan agriculture and careers. |
| **Key Inquiry Question** | Why is the concept of growth in plants important? |
| **Purpose in Storyline** | This final lesson closes the storyline by integrating all prior learning into a coherent, revised explanatory model that fully answers the driving question. Students revisit the Kisumu farmer phenomenon and demonstrate mastery through a performance task set in a Nakuru farming context. |
| **Safety Notes** | No significant hazards. If physical plant materials such as seedlings or stem cross-sections are brought in for reference, students should wash hands after handling soil or plant matter. Ensure students with plant allergies are seated away from samples. |

| B. LESSON OVERVIEW |
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| In this 40-minute synthesis lesson, students begin by briefly revisiting the anchoring phenomenon and their earliest predictions from Lesson 1 to surface how far their thinking has developed. They then individually revise and complete a labelled explanatory diagram tracing a seed from dormancy through germination, primary and secondary growth, hormone action, and environmental influences. The class reviews the Driving Question Board together, marking answered questions and celebrating the journey of inquiry. The lesson culminates in a structured performance task in which students explain a novel farming problem — pale, spindly bean seedlings in Nakuru — using all the concepts they have built. The lesson closes with a brief discussion connecting plant growth science to careers in agronomy, horticulture, and seed technology in Kenya. |

| C. LESSON IMPLEMENTATION FRAMEWORK | | | | |
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| **Phase** | **Learner Experience** | **Teacher Moves** | **Sensemaking Strategy** | **Formative Assessment Strategy** |
| **Predict Phase**<br>📹 **VIDEO:** [Balancing more complex chemical equations] | Students re-read their original Lesson 1 prediction cards or notebook entries — written when they first observed the maize and bean germination phenomenon | Teacher projects or reads aloud two or three anonymised Lesson 1 predictions that represent common early misconceptions, such as seeds failing to germinate because | Metacognitive annotation of prior predictions using a contrasting colour to make thinking growth visible. This activates prior knowledge and sets a reflective | Teacher circulates during the 3-minute silent annotation and notes which students still hold partial misconceptions — for example, conflating dormancy with seed |

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| Source: Khan Academy (English - US curriculum)<br>🔍 Search ARES for similar videos<br><br>📖 **HTML:**<br>Seed Dormancy and Germination - Advanced (Flexbook)<br>Source: CK-12<br>🔍 Search ARES for similar readings | and the dormant seed packet. Working individually for 3 minutes, they annotate their old predictions with a different colour pen, noting which predictions were confirmed, which were wrong, and which became more nuanced. They do not yet revise their full model; this is purely a metacognitive warm-up to surface the distance between their starting point and current understanding. | they are dead, or the idea that maize and beans emerge differently because of soil depth alone. Teacher says: In Lesson 1, many of us had ideas like these. Take 3 minutes in silence to look at your own first prediction and annotate it — what did you get right, what surprised you, what do you now understand more deeply? Use a different colour so we can see your thinking journey. Allow 3 minutes of silent individual work. Then invite two volunteers to share one thing they got right and one thing they had to change. Teacher affirms: That shift in thinking IS science. You built better models as you gathered more evidence, exactly the way agronomists and plant scientists at institutions like the Kenya Agricultural and Livestock Research Organisation work. | tone for the synthesis lesson. | death, or not connecting hormones to germination — to inform targeted prompting during the model revision phase. |
| **Observe Phase**<br>📹 **VIDEO:**<br>Balancing more complex chemical equations<br>Source: Khan Academy (English - US curriculum)<br>🔍 Search ARES for similar videos<br><br>📖 **HTML:**<br>Seed Dormancy and Germination - Advanced (Flexbook)<br>Source: CK-12<br>🔍 Search ARES for similar readings | Students observe a curated evidence gallery displayed around the classroom or on a shared board. The gallery contains: (1) a photograph of the Kisumu farmer garden showing bean seedlings with arched cotyledons above soil and maize with a single spike, (2) a cross-section diagram of a mango trunk showing annual growth rings, (3) a photograph of spindly sukuma wiki grown in low light alongside healthy sukuma wiki grown in full sun, (4) a data table from Lesson 3 showing germination rates under different water and temperature conditions, and (5) a simple diagram of auxin redistribution causing phototropism. Students spend 4 minutes doing a silent gallery walk, adding a sticky note to any image that connects to a concept they | Teacher sets up the gallery before the lesson. During the gallery walk, teacher circulates and uses targeted observation prompts without giving answers. At the mango trunk image, teacher may quietly ask a nearby student: What process explains the width of each ring you see here — primary or secondary growth? At the spindly sukuma wiki photograph, teacher may ask: Which hormone is most responsible for that stretching, and what environmental factor triggered its redistribution? Allow 4 minutes. After the gallery walk, teacher asks the class: Which image surprised you most — which one connects to something you had completely wrong in Lesson 1? Take 30 seconds to decide. WAIT TIME: 30 seconds of silence. Then take 2 rapid-fire | A silent gallery walk with sticky-note annotation encourages students to personally link visual evidence to scientific explanations and to make their growing competence visible to themselves and the teacher. | Teacher reads the sticky notes during transitions to assess which concepts — secondary growth, hormone action, dormancy causes — still have sparse or vague annotations, indicating areas to address during the explain phase and performance task. |

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| | can now explain that they could not explain in Lesson 1. | responses without discussion to maintain pace. | | |
| **Explain Phase**<br>📹 **VIDEO:**<br>Balancing more complex chemical equations<br>Source: Khan Academy (English - US curriculum)<br>🔍 Search ARES for similar videos<br><br>📖 **HTML:**<br>Seed Dormancy and Germination - Advanced (Flexbook)<br>Source: CK-12<br>🔍 Search ARES for similar readings | Students now complete the central task of the lesson: revising and finalising their individual explanatory diagram. Each student receives a blank A4 template with a horizontal timeline showing five labelled stages — Dormant Seed, Germination, Seedling with Primary Growth, Maturing Plant with Secondary Growth, and Reproductive Plant. For each stage, students must draw and label: the key biological structures active at that stage, the hormones involved and their specific actions, the environmental factors that support or limit that stage, and a connection arrow to the next stage explaining what triggers the transition. Students have 12 minutes to complete this individually. Teacher reminds them: Your diagram should be able to explain everything the Kisumu farmer saw — the different emergence of maize and beans, why some seeds in the packet failed to germinate, and why the mango tree trunk keeps getting wider every year. | Teacher begins the phase by saying: You now have all the pieces. Your job for the next 12 minutes is to put them together into one diagram that tells the complete story of how a plant grows — from a dormant seed all the way to a tree with a thick trunk or a branching herb. This diagram is your scientific model. Use everything from Lessons 1 to 7. WAIT TIME: Do not interrupt for the first 4 minutes. Circulate silently and observe. After 4 minutes, begin targeted verbal prompts to students who appear stuck: If a student has not connected hormones, ask: Which hormone controls whether the seed stays dormant or begins to germinate? What environmental signal causes that hormone to decrease? If a student has primary and secondary growth reversed, ask: Which type of growth happens at the tip of the shoot? Which one makes the trunk of a mango tree wider each year? With 2 minutes remaining, teacher says: Make sure your diagram can answer this question — why did the maize and bean seeds in Kisumu emerge so differently? If you can answer that with your diagram, you are ready. | The annotated timeline diagram is a cumulative explanatory model that requires students to integrate knowledge across all seven prior lessons. The explicit connection back to the anchoring phenomenon at the end of the instruction ensures the model serves a real explanatory purpose rather than being an abstract exercise. | Teacher collects the diagrams at the end of the lesson as the primary summative evidence of student learning. During the 12-minute work period, teacher uses circulating observations and targeted verbal prompts to identify and address persistent gaps in real time. Key things to look for: presence and accurate placement of all five hormones, correct distinction between epigeal and hypogeal germination at the Germination stage, and presence of vascular cambium at the Secondary Growth stage. |
| **Driving Question Board (DQB) Creation**<br>📹 **VIDEO:**<br>Balancing more complex chemical equations<br>Source: Khan | The class gathers around the Driving Question Board that has been displayed and updated since Lesson 1. The teacher reads aloud each remaining question from the What We Wonder column. Students vote by show of hands whether they can now answer that question confidently, partially, or | Teacher says: This board has been our inquiry map since Lesson 1. Every question on it was asked because of what that Kisumu farmer observed. Let us see how many we can now confidently answer. Teacher reads each question clearly and pauses for the vote. For answered questions, | Revisiting the DQB as a whole-class activity externalises the collective knowledge built over the unit and allows students to experience epistemic closure on the driving question while acknowledging the productive open ends of scientific inquiry. | The pattern of confident, partial, and unanswered votes across the class gives the teacher rapid formative data on which concepts are securely held by most students and which may need reinforcement in subsequent units or through teacher feedback on the collected diagrams. |

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| Academy (English - US curriculum)<br>🔍 Search ARES for similar videos<br><br>📖 **HTML:**<br>Seed Dormancy and Germination - Advanced (Flexbook)<br>Source: CK-12<br>🔍 Search ARES for similar readings | not yet. For questions with confident majority answers, a student volunteer gives a one-sentence explanation and the question is marked with a green tick. For partially answered questions, the class identifies exactly which part remains uncertain. The Driving Question Board review takes 5 minutes. Students also update the What We Now Know column with the three most important ideas from the whole unit, nominated by the class. | teacher prompts a student volunteer: Can you give us a one-sentence scientific explanation for that? WAIT TIME: 10 seconds after each question before asking for volunteers. Teacher celebrates growth explicitly: Look at how many questions this class has answered using evidence and investigation over eight lessons. That is what scientific inquiry looks like. For any question the class agrees is still partially open — perhaps around the detailed molecular mechanisms of hormone signalling — teacher validates: Science always leaves new questions open. That is not a failure; that is an invitation for further investigation. If you pursue biology, chemistry, or agronomy at university, those open questions are where research begins. | | |
| **Model Building Phase**<br>📹 **VIDEO:**<br>Balancing more complex chemical equations<br>Source: Khan Academy (English - US curriculum)<br>🔍 Search ARES for similar videos<br><br>📖 **HTML:**<br>Seed Dormancy and Germination - Advanced (Flexbook)<br>Source: CK-12<br>🔍 Search ARES for similar readings | In the final 8 minutes, students respond individually to the structured performance task in writing or, if the teacher chooses, in a brief oral explanation to a partner that the teacher assesses using a checklist. The performance task prompt is: A farmer in Nakuru notices her bean seedlings are growing tall and pale instead of short and bushy. Using what you know about plant growth and development, explain what might be happening and what she could do. Students are expected to reference: the role of light as an environmental factor, auxin redistribution causing etiolation and elongation in low-light conditions, the distinction between primary growth in length versus the absence of lateral branching due to apical dominance, and a practical recommendation such as | Teacher introduces the performance task by saying: Here is your final challenge — a real farming problem set in Nakuru. I want you to act as a plant scientist advising a farmer. Use everything from this unit. You have 6 minutes to write your explanation. WAIT TIME: Allow 6 minutes of uninterrupted individual writing. Do not prompt or hint during this time. After 6 minutes, teacher invites one or two volunteers to share their explanation briefly. Teacher gives affirming and specific feedback: I heard you connect low light to auxin redistribution — that is exactly right. I also heard you recommend increasing spacing to allow more light in — that is a practical, evidence-based recommendation a real agronomist would make. Teacher closes with the career connection, saying: | The performance task requires transfer of integrated knowledge to a novel, Kenya-specific agricultural scenario, ensuring that learning is not merely reproductive but genuinely explanatory and applicable. The career connection closes the loop between school science and real-world relevance in the Kenyan context. | The written performance task response is the primary summative assessment tool for this lesson and the unit. Teacher assesses responses using a 4-point rubric: (1) identifies light as the environmental factor involved, (2) correctly names and explains auxin redistribution or etiolation, (3) connects the symptom to primary growth and apical dominance mechanisms, and (4) provides a scientifically sound and practical recommendation. Oral responses are assessed using the same rubric via a checklist during pair sharing. |

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| | increasing light exposure, spacing seedlings, or pinching the apical bud to promote lateral branching. After completing the performance task, the teacher closes with a brief 2-minute career connection, naming specific Kenyan careers and institutions: agronomists at KALRO in Nairobi advising on optimal planting conditions, horticulturalists in Thika managing flower and vegetable production for export, seed technologists testing dormancy-breaking methods for the Kenya Seed Company in Kitale, and researchers at the University of Nairobi Department of Botany studying plant hormones for crop improvement. | Every concept you used today — dormancy, germination, primary growth, hormones, environmental factors — is used every day by Kenyan scientists and farmers to grow better crops, develop new seed varieties, and feed our country. You are thinking like scientists. | | |

## D. TEACHER REFLECTION

After the lesson, consider the following questions: Did students demonstrate in their diagrams that they could integrate germination conditions, growth types, hormone functions, and environmental factors into a coherent explanatory model, or did their diagrams treat these as separate, disconnected topics? Were students able to apply integrated knowledge to the Nakuru farmer performance task without prompting — specifically connecting low light to auxin redistribution and etiolation? Did the DQB review reveal any concepts that a majority of students still hold as misconceptions — such as conflating dormancy with seed death, or attributing secondary growth to apical meristems? Which students remained silent or produced very sparse diagrams, and how will you provide follow-up support through feedback on their collected work? Did the career connection land as meaningful and relevant, or did it feel rushed — and if so, how could it be woven earlier into the unit in future iterations? Consider whether the 40-minute period was sufficient for both the model revision and the performance task, and adjust the time allocation between the gallery walk and the diagram completion if needed when teaching this unit again.

## E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

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| **What did I observe?** | We observed the complete evidence gallery from the unit — maize and bean germination in the Kisumu garden, the mango trunk cross-section with growth rings, spindly sukuma wiki grown in low light versus healthy sukuma wiki in full sun, germination data tables, and auxin redistribution diagrams — and we revisited the Driving Question Board to see every question our class has answered since Lesson 1. |
| **What did I learn?** | We learned that a seed journey from dormancy to a mature plant is controlled by interacting biological and environmental systems: dormancy is maintained by abscisic acid and broken by water, temperature, and scarification; germination type — epigeal in beans and hypogeal in maize — is determined by the position of cotyledons; primary growth at apical meristems increases length while secondary growth at the vascular cambium increases girth; auxins, gibberellins, cytokinins, abscisic acid, and ethylene each regulate specific stages; and environmental factors including water, temperature, light, and nutrients interact with these hormones |

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| | to produce the plant forms seen across Kenyan regions from Kericho to Mombasa. |
| **How does this explain the phenomenon?** | We explained the complete anchoring phenomenon: bean seeds in Kisumu undergo epigeal germination because the hypocotyl elongates and pushes cotyledons above soil, while maize undergoes hypogeal germination because the epicotyl elongates and the coleoptile emerges while cotyledons remain underground; stored seeds failed to germinate because dormancy was maintained by abscisic acid and unfavourable conditions; and the mango trunk widened over the years through secondary growth driven by the vascular cambium. We also explained the Nakuru farmer phenomenon — pale, spindly bean seedlings result from etiolation caused by auxin redistribution in low light, which promotes excessive primary growth in length while suppressing lateral branching through apical dominance, and the solution is to increase light exposure or remove the apical bud to break apical dominance. |

| DIFFERENTIATION AND INCLUSION | | |
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| **Learner Need** | **Support Strategies** | **Extension Strategies** |
| English Language Learners | Provide bilingual glossaries; allow use of first language for initial model drawing. | Write extended explanations of observations; lead group discussions in English. |
| Students with learning difficulties | Provide structured note frames; pair with a supportive partner during investigations. | Mentor peers; create additional visual models to explain concepts. |
| Gifted and advanced learners | Access to additional reading materials; challenge questions during DQB. | Lead model-building discussions; research and present extension topics to class. |